# Term 3: Electrical | Electronic systems

## Chapter 11
# Component symbols and simple circuits

In this chapter, you will revise the work you did on electrical systems and control in Grade 8. You will also revise simple circuits, circuit diagrams and connecting cells, and lamps and switches in series and parallel. You will then do action research on the effects of changing the voltage in a circuit.




*Figure 1: A torch and batteries*

## 11.1 Revision 1: Component symbols

**Components** are the parts that we connect in an electric circuit.

Do you remember the symbols for cells, lamps and switches?

Do you remember the difference between joining components in series and in parallel? Let's see what you can remember.

You have already learnt that an electric circuit is a closed path in which a current flows.

The simplest circuit has:
- a power source such as a cell,
- a conductor, and
- a load that provides resistance, such as a lamp.

### Cells in series

**In series** means the cells are connected end-to-end, and the current flows through each cell in turn.

Two or more cells can be connected **in series** to increase the voltage in the circuit. Figure 2 below shows two cells connected in series in a circuit. The positive terminal of cell A is connected to the lamp. The negative terminal of cell A is connected to the positive terminal of cell B, and the negative terminal of cell B is connected to the other terminal of the lamp.

1. Draw a circuit diagram of the circuit in Figure 2.

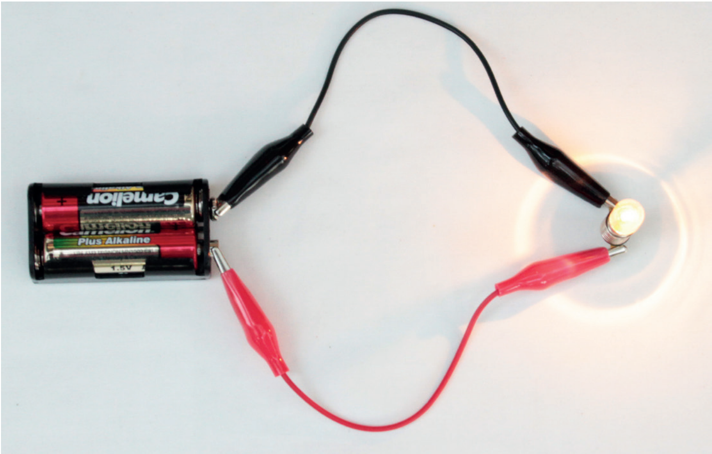



*Figure 2: Two cells in series connected to a lamp*

2. Figure 3 below shows three cells connected in series in a circuit. Draw a circuit diagram of the circuit.

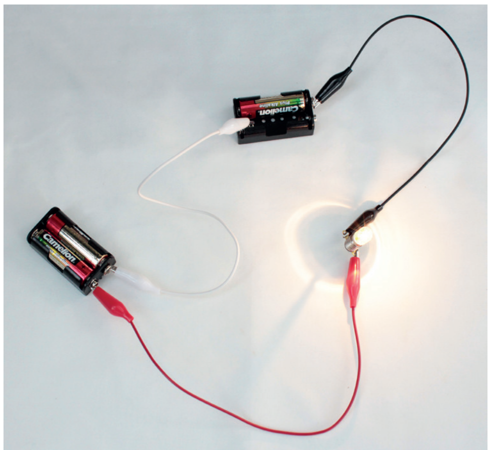


*Figure 3: Three cells in series connected to a lamp*

When cells are connected in series, their total voltage is the sum of the voltages of the three cells:
1,5 V + 1,5 V + 1,5 V = 4,5 V

## Cells in parallel

Two or more cells can also be connected **in parallel**. A parallel circuit has two or more different paths for the current to travel along.

Figure 4 below shows three cells connected in parallel in a circuit. The positive terminals of all three cells are connected to one another and to the lamp. The negative terminals of all three cells are connected to one another and to the other terminal of the lamp.

1. Draw a circuit diagram of the circuit in Figure 4.

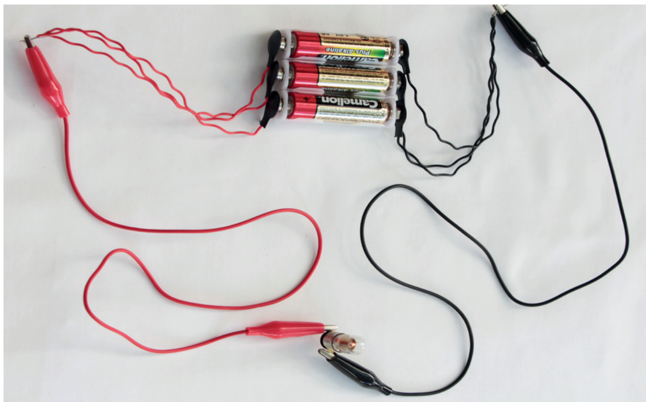


*Figure 4: Three cells in parallel connected to a lamp*

When cells are connected in parallel, the total voltage of the cells is the same as that of a single cell (1,5 volts)

## Lamps in series

Two or more lamps can also be connected in series.

The pictures below show circuit diagrams of two and three lamps connected in series with the battery. The positive terminal of the battery (B+) is connected to lamp 1, the other side of lamp 1 is connected to lamp 2, the other side of lamp 2 is connected to the negative terminal (B–) of the battery, and so on.

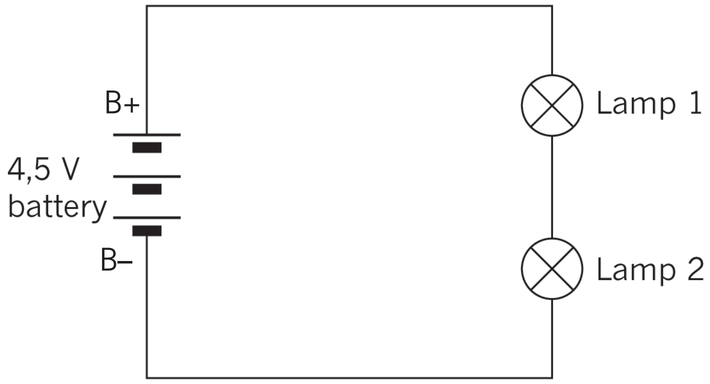



*Figure 5: Two lamps in series*

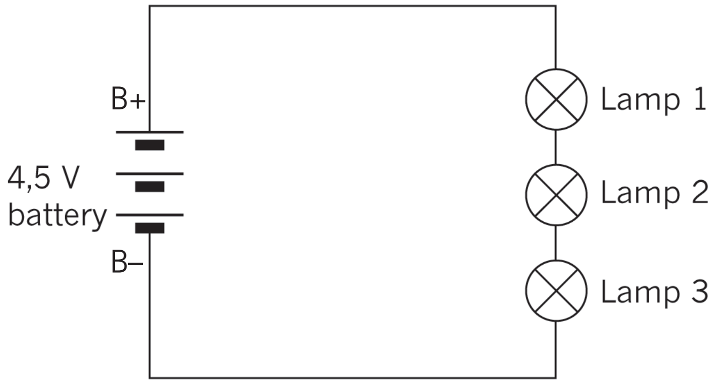



*Figure 6: Three lamps in series*

1. How does increasing the number of lamps in series change the current and voltage in the circuit?

If all the lamps have the same resistance, the voltage drop across each lamp will be equal to 1,5 V.
When the voltage drops of all the lamps are added, the total battery voltage of 4,5 V is obtained.
The current is the same through each lamp.

## Lamps in parallel

Two or more lamps can also be connected to the battery in parallel, as shown in the pictures below. The positive terminal of the battery is directly connected to one side of each lamp and the negative terminal to the other side of each lamp.

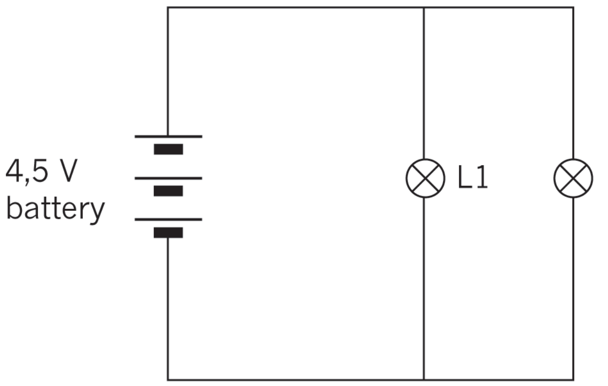



*Figure 7: Circuit diagram of two lamps in parallel*

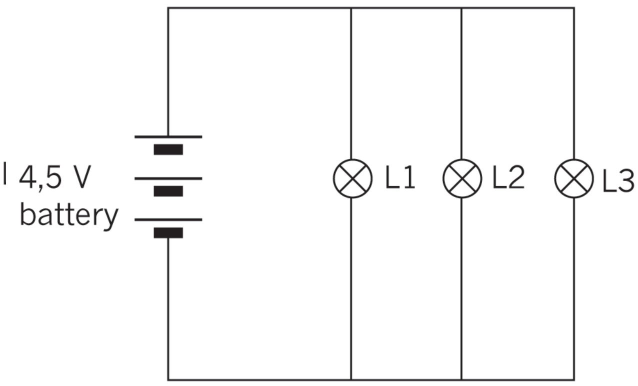



*Figure 8: Circuit diagram of three lamps in parallel*

> The applied voltage is the same across each lamp.
> The current splits so that some current goes through each lamp. If we add the three currents in the lamps we find the total current from the battery: $I_t = I_1 + I_2 + I_3$

1. Look at the circuit diagram in Figure 9 and answer the following questions:

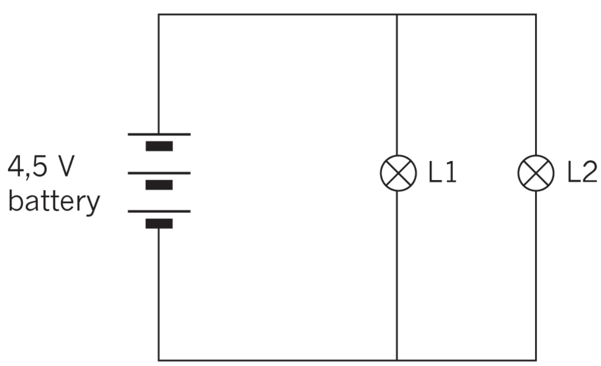



*Figure 9*

(a) What is the voltage drop across lamps 1 and 2?

(b) The total current in the circuit is 10 A. If lamp 1 has a current of 4 A flowing through it, what will the current be through lamp 2?

## Switches in series and parallel

In a circuit with one switch, the switch controls whether the current flows through the circuit or not. If the switch is open, no current flows, as the circuit is not completed. The closed switch allows the current to flow.

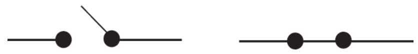


*Figure 10: Symbols for an open switch and a closed switch*

We can use two or more switches to control components in a circuit in more complex ways.

> In a logic circuit, an open switch is regarded as having a value of 0, and a closed switch as having a value of 1.
> The switches are the inputs that control the final state of the circuit.
> If the circuit is not completed, the output is in the OFF state and has a value of 0.
> If the circuit is completed, the output is in the ON state and has a value of 1.

### Switches in series

In the circuit below, there are two switches in series. This gives us four different switch combinations. They are:

- switch A and B both open,
- switch A open and B closed,
- switch A closed and B open, and
- both switches closed.

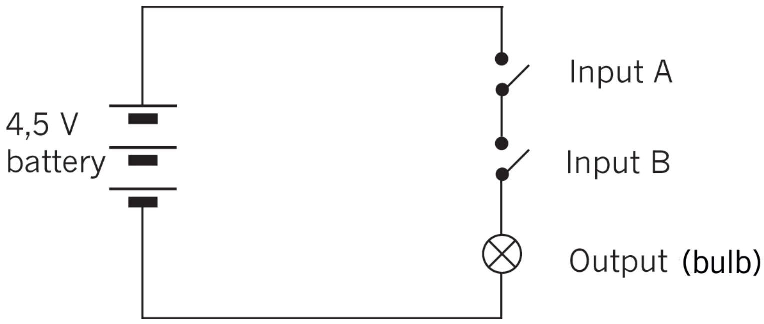



*Figure 11: Circuit with two switches in series*

Do you see that the current cannot flow through the circuit if either switch A or switch B is open? Both of them must be closed for the lamp to glow.

1. In the table below, "0" means off or open, and "1" means on or closed. Copy and complete the table to show all the different combinations possible in the circuit in Figure 11. To help you, the first two rows of the table have already been completed. Make sure you understand those two rows before you complete the table.

| **Input A** | **Input B** | **Output** |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | |
| 1 | 1 | |

The table showing these combinations is called a **truth table**. Both switch A *and* switch B must be closed for the circuit to be completed (an output of 1).

So we can see that switches connected in series give us an **AND** function.

## Switches in parallel

In the circuit below, there are two switches in parallel. This also gives us four different switch combinations.

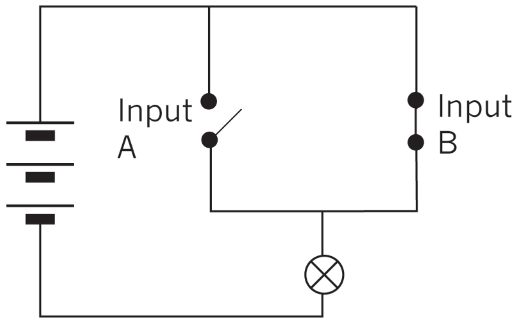



*Figure 12: Circuit with two switches in parallel*

Do you see that the current can go through the closed switch, even if the other switch is open?

1. Draw up a truth table for the circuit in Figure 12.

The truth table shows that when switch A *or* switch B is closed, the output will be 1 (the lamp will be on).

We call switches in parallel an **OR** function.

## Questions for homework

1. Would the lamp glow in each of these circuits? Explain your answer.

(a)

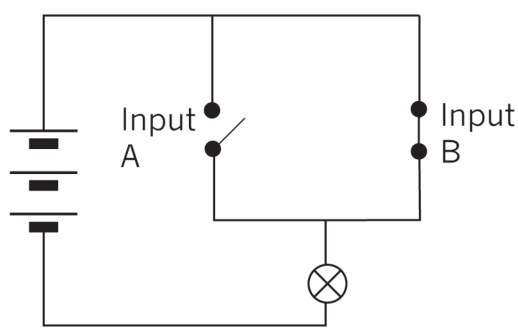



*Figure 13*

(b)

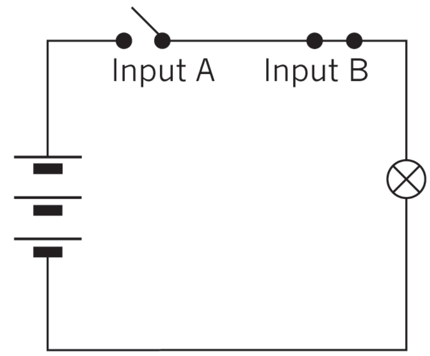



*Figure 14*

(c)

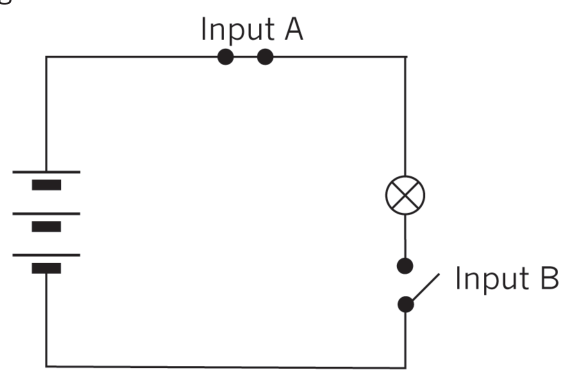



*Figure 15*

2. A kettle must be switched on at the wall plug first and then at the kettle itself.

(a) Copy the truth table below. Fill it in to show all the possible combinations.

| Wall plug switch | Kettle switch | Output |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

(b) Is this an AND function or an OR function? Explain your answer.

## 11.2 Revision 2: Simple circuits

In this lesson, you will set up simple circuits, revising what you learnt about setting up circuits in Grade 8.

### Set up circuits

**You will need the following for this activity:**

- two AA cells in cell holders,
- connecting wires,
- a switch, and
- two lamps.

Note that you can use a homemade switch and a cell holder made of insulation tape for this activity.

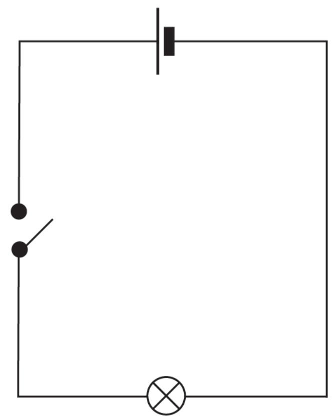


*Figure 16*

1. Look at the circuit diagram on the right.
   Set up this circuit and check that it works, by closing the switch.
   Does the lamp glow?

When you have the circuit working correctly, move on to question 2.

If you need to, you can troubleshoot your circuit by looking at the following:

- If the lamp doesn't light up, but the wires get hot, you may have a short circuit. This means that the lamp is not connected correctly in the circuit, or that it is faulty. Check that the lamp is connected correctly in the circuit.
- If the lamp still doesn't light up, check each component and connecting wire by replacing them, one by one. You can identify which one is faulty this way.

2. Add another lamp to the circuit in series with the first one.
   (a) Draw a circuit diagram for this circuit.
   (b) What do you notice about the brightness of the lamps?
3. Set up the same circuit, but add another bulb in series with the first bulb.
   (a) Draw a circuit diagram for this new circuit.
   (b) Write what you notice about the lamps in this circuit.
4. Write down your conclusions about changing the number of cells and the number of lamps in the circuit.

## 11.3 Testing voltage and current in circuits (Ohm's Law)

In this lesson, you will investigate the relationship between the values of the voltage and the current in a circuit. You will need to use a multi-meter that can be set to measure the voltage, resistance or the current in a circuit.

**V**: volts (potential)
**A**: amps (current)
**Ω**: ohms (resistance)

Begin by reading the text below on how to use a multi-meter correctly.

### Measuring resistance

Identify the section labelled "Ω" on the multi-meter dial.

- Connect the red test lead to the "V Ω mA" terminal, and the black test lead to the "COM" terminal.
- Adjust the function selector switch to the highest range in the section labelled "Ω". There are different resistance ranges on a multi-meter, such as 200 Ω, 2 kΩ, 20 kΩ, 200 kΩ and 2 MΩ.
- Connect the ends of the test leads across the unknown resistor as shown. Ensure that the resistor is isolated from any other component or power supply.
- Read the resistance value from the display. If the displayed value is zero or very small, for example 0,001, then switch the function selector switch to a lower resistance range. Keep on switching to lower resistance ranges until the displayed value is larger than 1. Remember that the displayed value is for the units of measurement indicated by the range you selected. It may be Ω, kΩ or MΩ.

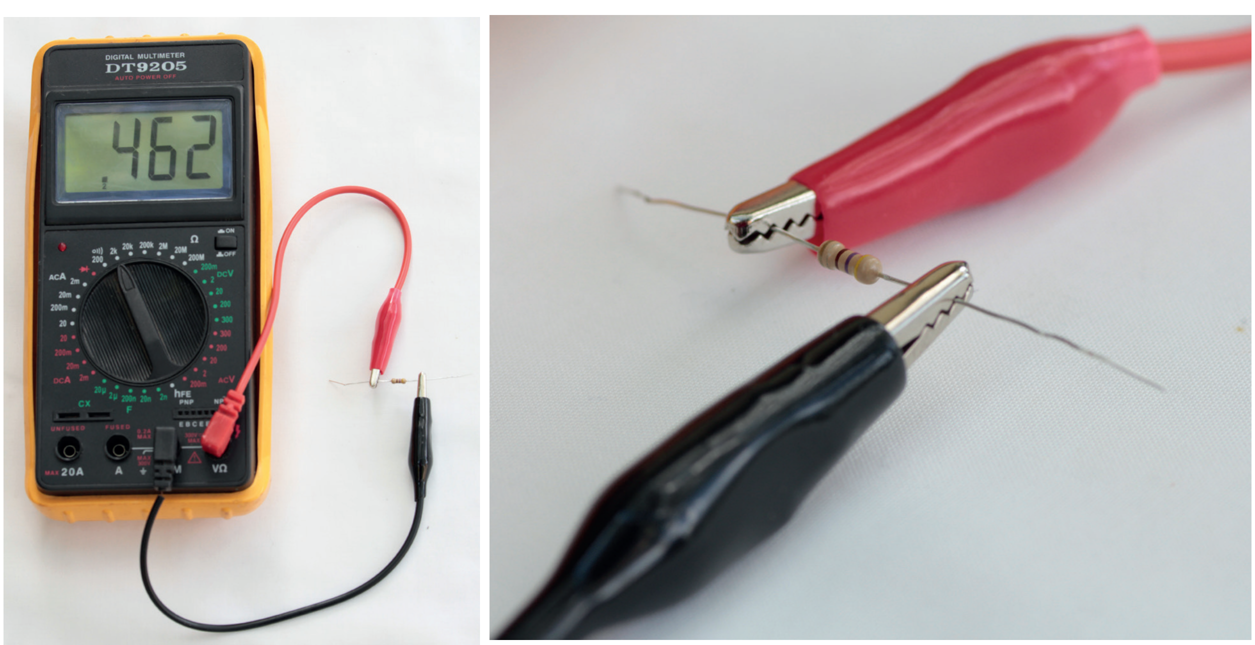



*Figure 17: Multi-meter set and connected to measure resistance*

## Measuring voltage

Identify the section labelled “DCV” on the multi-meter dial.

- Connect the red test lead to the “V Ω mA” terminal, and the black test lead to the “COM” terminal.
- Adjust the range selector to the “DCV”.
- Set the meter on the highest range.
- Connect the other ends of the test leads parallel across the part of the circuit where the voltage is to be measured: red test lead to positive (+), and black test lead to negative (–).
- Read the voltage from the display. You may need to adjust the function selector to choose a different voltage range, so that the reading is displayed properly and accurately.

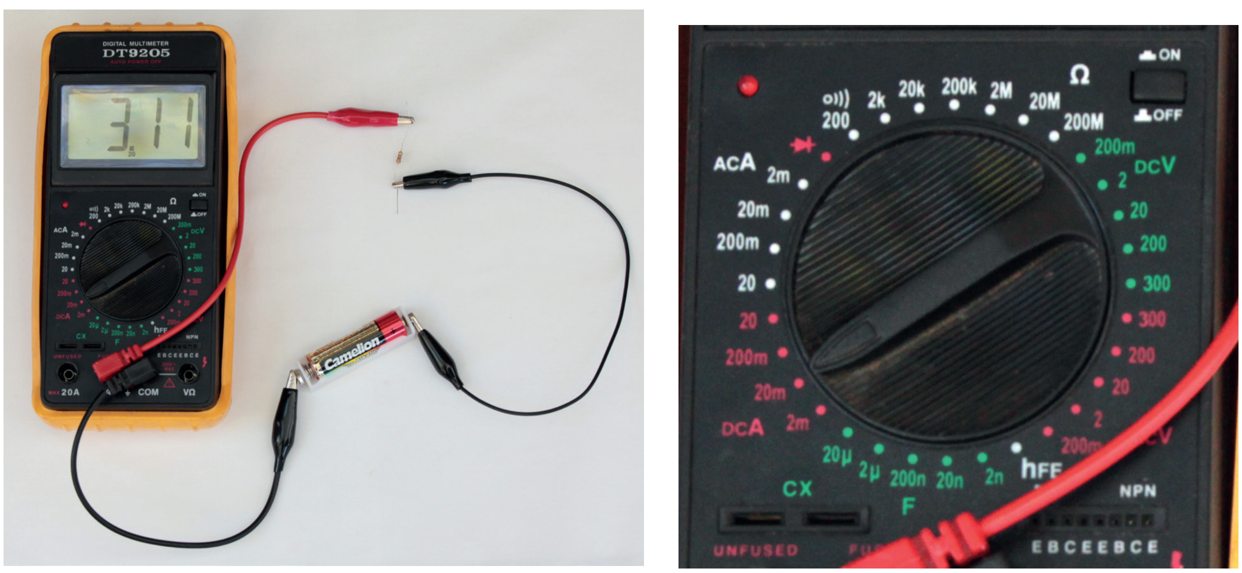



*Figure 18: Multi-meter set and connected to measure current*

## Measuring current

Identify the section of the dial labelled “DCA” on the multi-meter dial.

- Connect the red test lead to the “V Ω mA” terminal and the black test lead to the “COM” terminal. If the current to be measured is between 200 mA and 10 A, connect the red test lead to the “10 A” terminal.
- Adjust the function selector to the “A” (ampere) region. If you are measuring an unknown current, start from the highest range, then adjust to a proper lower range for the best accuracy.
- Connect the other ends of the test leads in series with the part of the circuit where the current is to be measured. (Disconnect the circuit and place the meter in series.)
- Read the current value from the display.

## Action research

**You will need the following for this activity:**

- three penlight cells (AA) in holders,
- a 500 ohm resistor, with the colour bands exactly as in Figure 19, and
- two multi-meters, or an **ammeter** and a **voltmeter**.

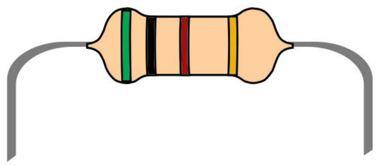


*Figure 19: A 500 ohm resistor*

1. Set up the circuit as shown in Figure 20 below, using a cell, resistor and ammeter. If you use a multi-meter instead of an ammeter, set it on the amps scale.

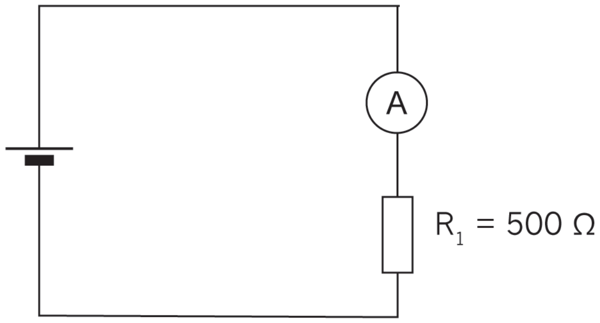



*Figure 20: Circuit with one cell, resistor and ammeter*

2. Now connect a voltmeter across the resistor, as shown in Figure 21. If you use a multi-meter instead of a voltmeter, set it on the volts scale.

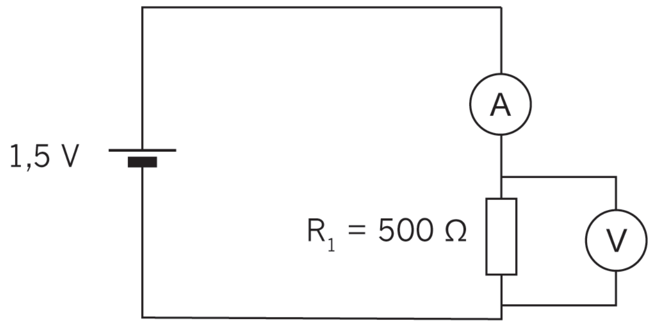



*Figure 21: Circuit with one cell, resistor, ammeter and voltmeter across resistor*

Record the readings.

In the next chapter, you will learn how the colour bands on a resistor tell you the resistance (ohms).

An **ammeter** is always connected in series with the part of the circuit for which you measure the current, so that it measures the full current through that part of the circuit. It has a very small resistance so that it does not change the current in the circuit.

A **voltmeter** is always connected in parallel with the part of the circuit for which it measures the voltage between two points. Very little current flows through a voltmeter since it has a very high resistance.

3. Now connect a second cell in series as shown in the circuit diagram below:

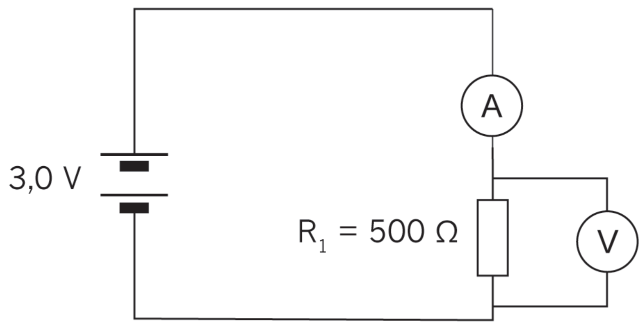



*Figure 22: Circuit with two cells in series, resistor, ammeter and voltmeter across resistor*

Record the readings.

4. Now connect a third cell in series as shown in Figure 23.

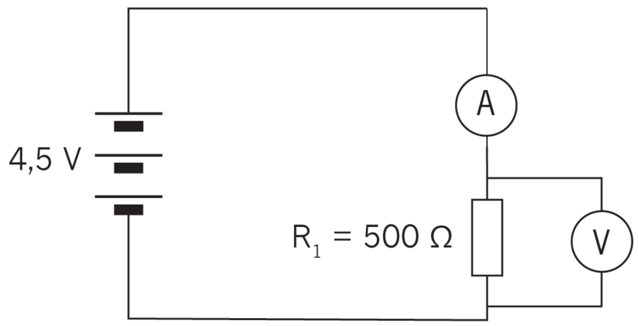



*Figure 23*

Record the readings.

5. Copy the table below and fill in your readings.

| | **With one cell** | **With two cells** | **With three cells** |
|---|---|---|---|
| **Voltage** | | | |
| **Current** | | | |

6. Copy the set of axes below onto graph paper. Plot the graph.

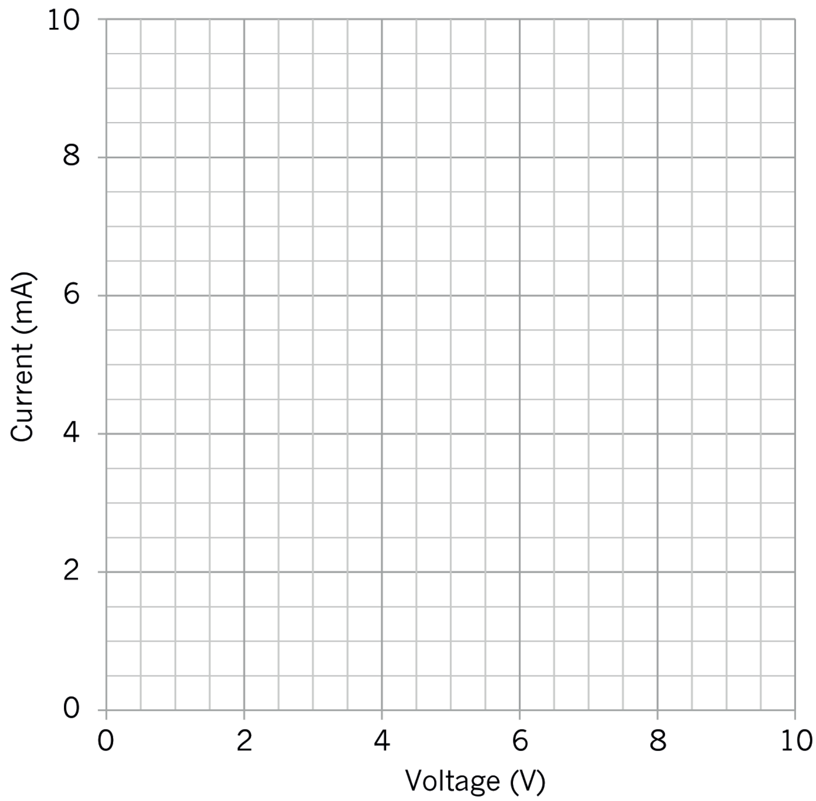



*Figure 24*

7. Describe the relationship between voltage and current for a 500 Ω resistor.

- Did you notice that as the voltage is increased the current increases?
- Is your graph in a straight line?

There is a **directly proportional relationship** between voltage and current. As the voltage is doubled, the current will double; and as the voltage is tripled, the current will triple.

## Next week

Next week, you will look at different kinds of resistors used in circuits. You will also practise doing calculations using the formulas in Ohm's Law.

# Chapter 12
# Resistor colour codes and Ohm's Law

In this chapter, you will learn how to use resistors in electric circuits to control a current. You will discover that there are different kinds of resistors for different purposes, and you will learn how to read the amount of resistance on a resistor. You will also learn about Ohm's Law, which relates the quantities of voltage, current and resistance, and you will use formulas to calculate the values of voltage, current and resistance.



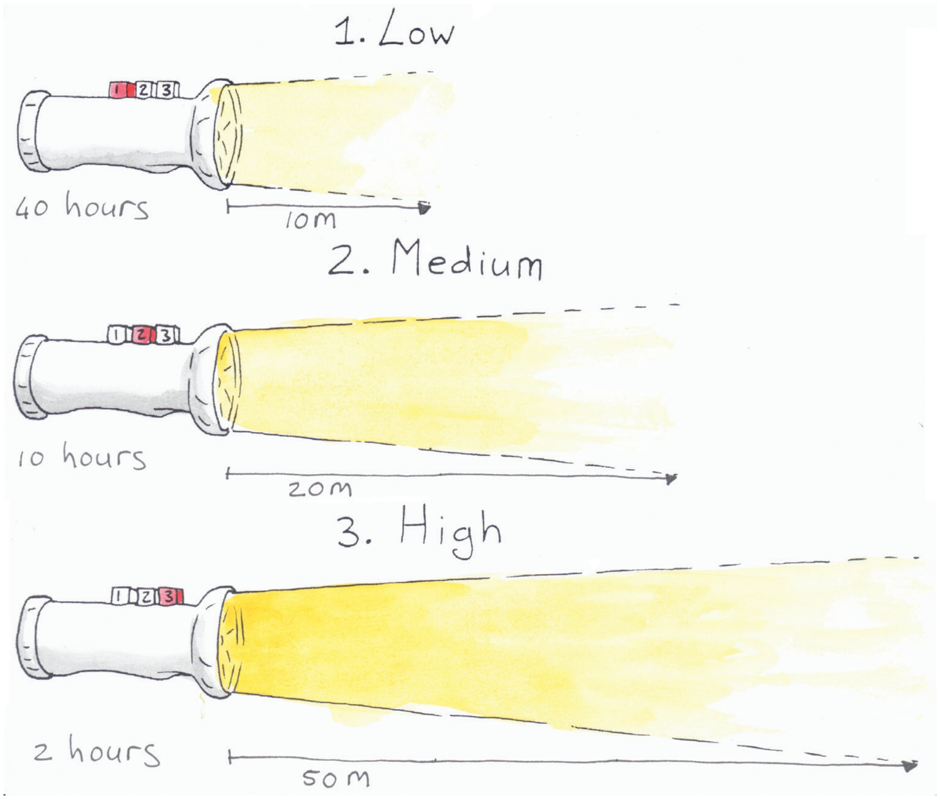



*Figure 1: You can change the brightness of the light on some torches. The brighter the light you choose, the faster the battery will run out.*

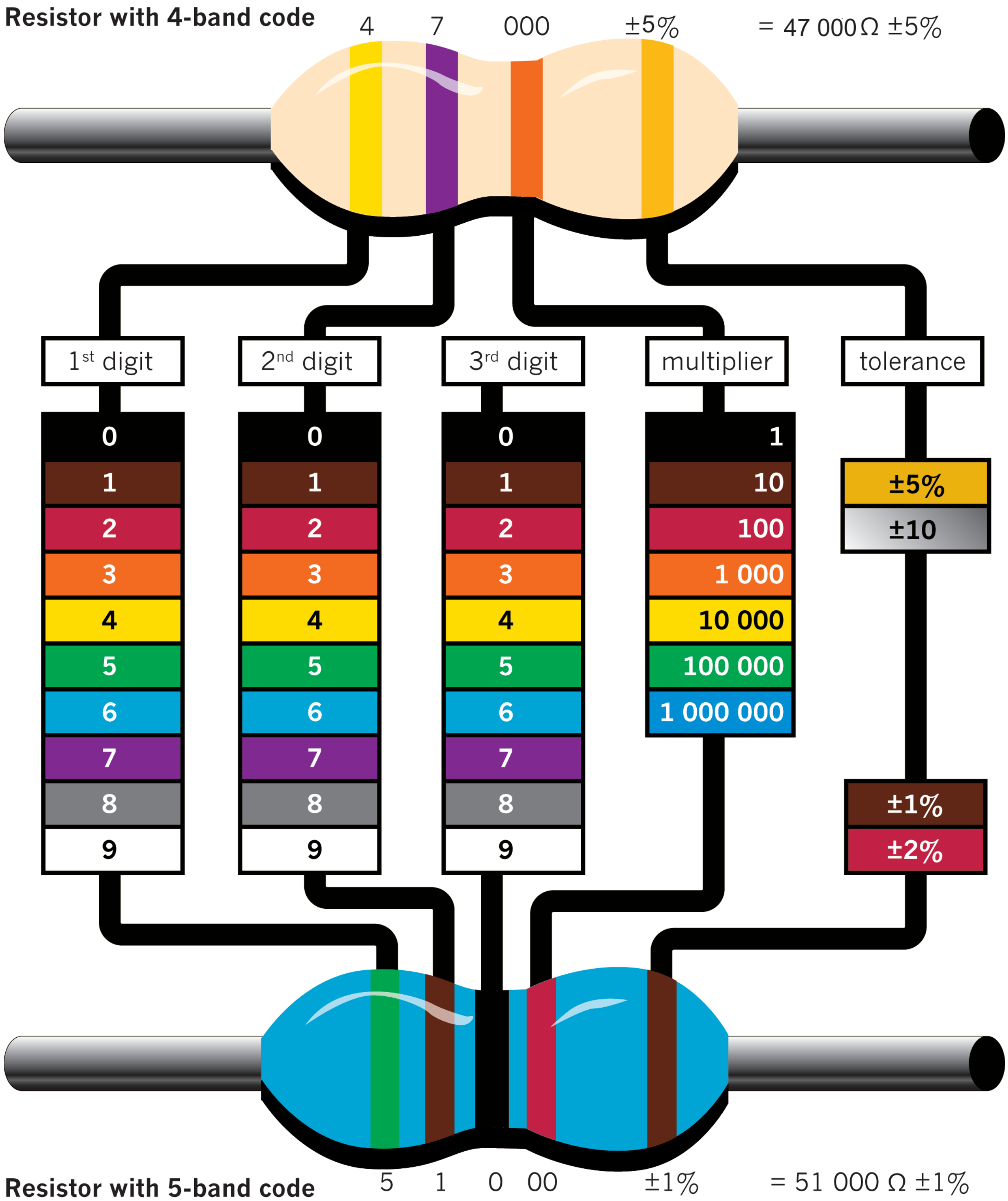



*Figure 2: How to read the colour bands on a resistor to find out what its resistance is. (You will only work with resistors with four-colour bands, such as the one at the top.)*

## 12.1 Resistors and their identification codes

### What is resistance?

Electricity flows far more easily through copper wire than through plastic wire, string or grass. Copper wire has a low resistance to electricity flow, whereas plastic wire has a high **resistance**. Because electricity flows easily through copper wire, copper is a good **conductor** of electricity.

The resistance that an object, for example a piece of wire, offers to the flow of electricity can be measured.

> Resistance is measured in ohms. We use the symbol Ω.

To **resist** something means to try prevent it. If you sit in a tree and the wind blows hard, you can resist falling down by clinging to the branches.

To **conduct** means to allow something to pass through.

When electricity flows through a conductor, heat is generated. Some metals, such as nickel and chrome, resist the flow of electricity quite strongly, and heat up when even a small electrical current is forced to flow through it. The heating elements of stoves and kettles are normally made of a mixture of nickel and chrome. When some metals get extremely hot, they **emit** light.

When something **emits** light, it is a source of light. A light bulb is a source of light, but a mirror is not a source of light as it only reflects light.

If the resistance in a circuit is very low, for example when the terminals of a cell are connected with a piece of thick copper wire, the current will flow very strongly. This is called a "short circuit". It can result in so much heat being generated that damage is caused to the cell and other parts of the circuit, the conducting wires can melt and a fire can start.

By adding more resistance to a circuit, you can control how great the current is that flows through the circuit. In this way, you can protect the components in a circuit from too much current flowing through them. Increasing the resistance also means the cell or battery powering the circuit will last longer. You can add precise amounts of resistance by using resistors with the required resistance value.

## What is a resistor?

A resistor is a specially designed component that is normally used in a circuit to limit the current. Resistors are made of materials with a high resistance to electricity flow, and come in the form of thin wires or films. Resistors also have precise resistance values that don't change much in different environmental conditions.

The most commonly used resistors look like tubes, with two wires to connect it to the circuit. The symbol to show a resistor in a circuit diagram is an open rectangle or a zigzag line.

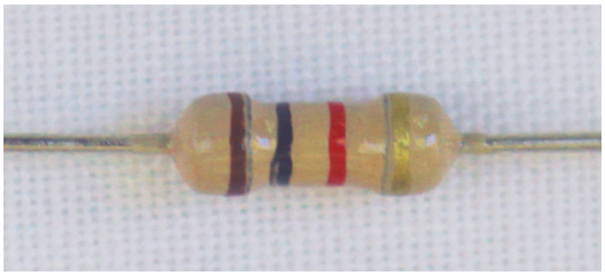


*Figure 3: A typical resistor*

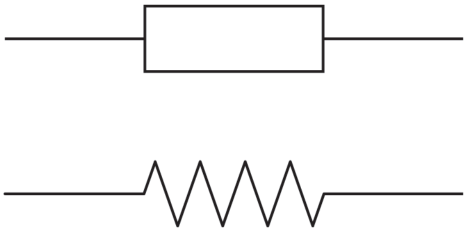


*Figure 4: Circuit symbols for resistors*

Low-value resistors often have their resistance value printed on them in numbers, while high-value resistors are coded, using coloured bands. The first three bands give the value of the resistor in ohms. The colour-code chart on the second page of this chapter will help you to work out the resistance value in ohms.

Resistors are the most commonly used components in electronics, as they are useful to control current. You will see how they are used in the following weeks.

The fourth band on a resistor shows the accuracy rating as a percentage. This is also called the "tolerance". The band is gold or silver, depending on the tolerance. For the circuits you will be building, this is not important.

**Units of measurement: ohms, kilo-ohms and mega-ohms**

- 1 kΩ = 1 000 Ω = $10^3$ Ω
- 1 MΩ = 1 000 kΩ = 1 000 000 Ω = $10^6$ Ω

**Kilo** means multiply by a thousand, for example 1 km = 1 000 × 1 m.

**Mega** means multiply by a million.

## The resistance of resistors

1. Work out and write down the resistance of each of these resistors:

(a)

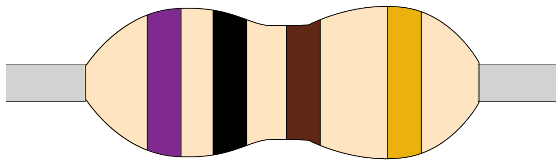


*Figure 5*

(b)

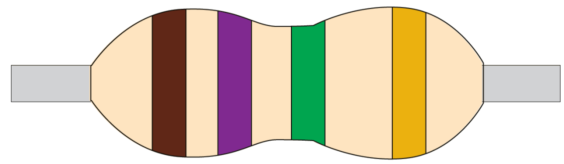


*Figure 6*

(c)

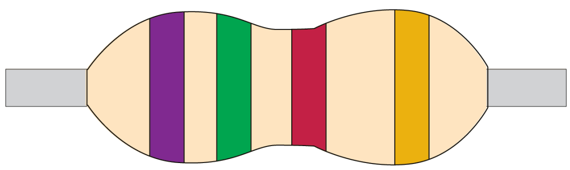


*Figure 7*

(d)

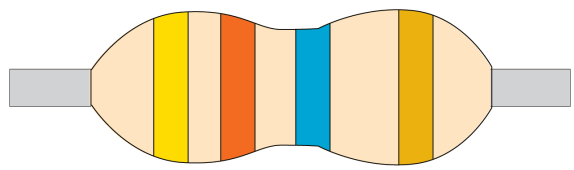


*Figure 8*

2. Copy blank resistors with the bands into your books. Fill in the colour codes to show the given resistance. If you don't have coloured pencils or pens, write the colour of each band above it.

(a) 200 kΩ

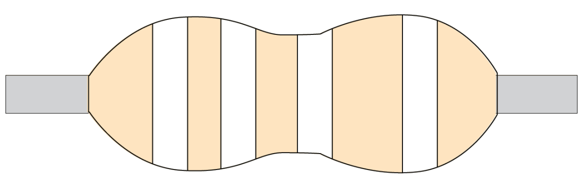


*Figure 9*

(b) 300 Ω

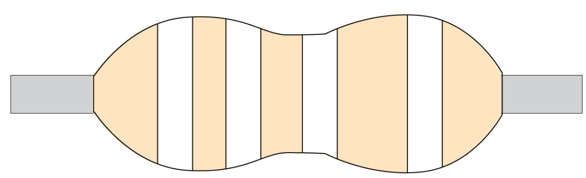


*Figure 10*

3. Describe the function of a resistor as a component in an electrical circuit.

## 12.2 Ohm's Law

There is a special relationship between the voltage, current, and resistance in any circuit. You can control any one of these three **variables** by changing the other two variables.

Ohm's Law states that as voltage increases, the current increases if the resistance is **constant**.

A **variable** is a quantity that can have different values, for example the amount of water in a tank. A **constant** is a quantity that always has the same value, for example gravitational acceleration. Sometimes we call a quantity a constant because we decide to keep it constant.

In the formula for Ohm's Law:

- **V** is the **potential** or **voltage difference** measured in volts,
- **I** is **current** measured in amps, and
- **R** is **resistance** measured in ohms.

Figure 11 shows this relationship in a formula triangle.

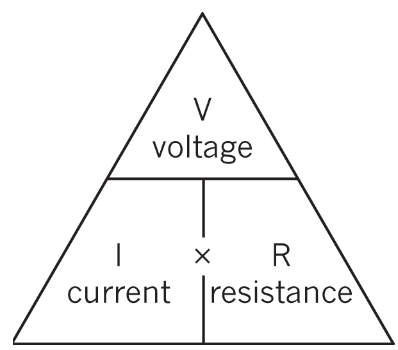



*Figure 11*

When the voltage and current are known, the resistance can be calculated with:

$$R = \frac{V}{I}$$

When the resistance and current are known, the voltage can be calculated with:

$$V = I \times R$$

When the resistance and voltage are known, the current can be calculated with:

$$I = \frac{V}{R}$$

## Questions

Consider the following circuit in Figure 12 on the right:

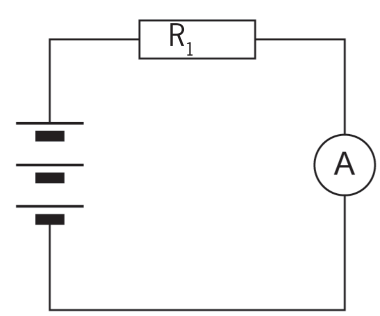



*Figure 12*

1. What does Ohm's Law say will change in a circuit when the resistance is kept constant but the number of cells in series is increased?
2. How will the current change if the voltage supplied by the battery of cells is kept constant but the resistor is replaced by another resistor with a lower resistance?

3. How would you describe the relationship between the current and the voltage in a circuit?
4. Which of these changes will cause the current through an electrical circuit to decrease? Write down all the letters of the statements that are correct.
   (a) a decrease in the voltage
   (b) a decrease in the resistance
   (c) an increase in the voltage
   (d) an increase in the resistance
5. An electrical circuit has three 1,5 V cells in series that is connected to a lamp and a resistor in series. Which of the following things would cause the lamp to shine less brightly? Write down all the letters of the statements that are correct.
   (a) an increase in the voltage of the battery (add another cell)
   (b) a decrease in the voltage of the battery (remove a cell)
   (c) a decrease in the resistance of the resistor
   (d) an increase in the resistance of the resistor

## 12.3 Calculations using Ohm's Law

In the previous lesson, you learnt how Ohm's Law can be used to predict what will happen when you change one or two of the following variables: current, voltage or resistance. You will now use the formulas of Ohm's Law to make predictions. Remember to use the correct units in the formula!

**Example 1**

Calculate the value of the resistance in the diagram below, if the voltage across the resistor is 12 V and the current through the resistor is 2 A.

$$R = \frac{V}{I}$$
$$= \frac{12\ V}{2\ A}$$
$$= 6\ \Omega$$

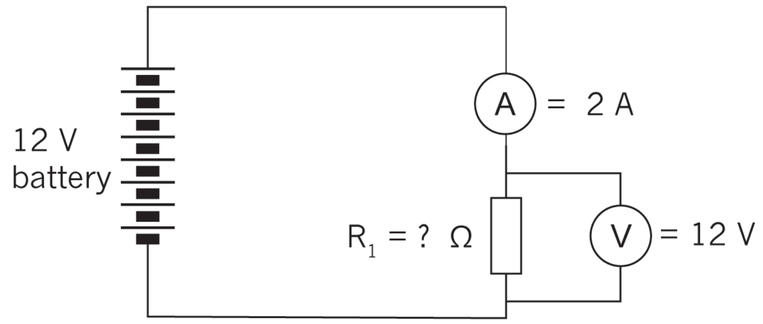



*Figure 13*

**Example 2**

Calculate the value of the voltage supply in the circuit below if the resistor has a value of 4 Ω and the current through the resistor is 2,5 A.

$V = I \times R$

$= 2{,}5\ \text{A} \times 4\ \Omega$

$= 10\ \text{V}$

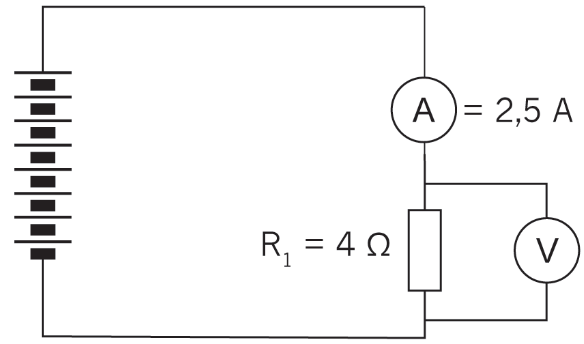



*Figure 14*

**Example 3**

Calculate the value of the current in the circuit below if the resistor has a value of 3 Ω and the voltage across the resistor is 12 V.

$I = \frac{V}{R}$

$= \frac{12\ \text{V}}{3\ \Omega}$

$= 4\ \text{A}$

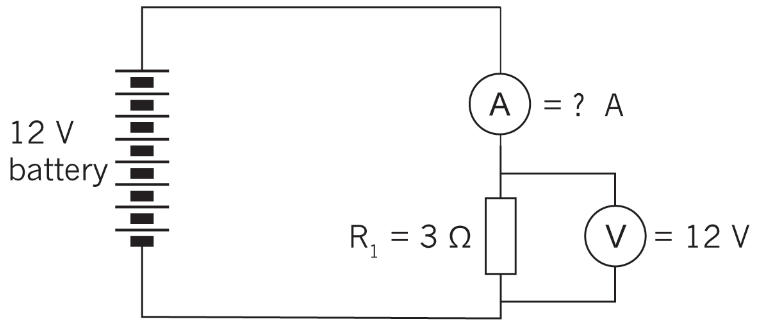



*Figure 15*

## Questions

1. What will the potential difference be if the current in a circuit is 10 A and the total resistance is 1 000 Ω?
2. Given $V$ = 10 V and $R$ = 1 kΩ, what will the value of the current be in a circuit?
3. Given $V$ = 20 V and $R$ = 5 kΩ, solve for the current.
4. A tumble dryer in a laundry service uses a 220 V power source. The coils of the heater provide an average resistance of 12 Ω. What is the current flowing through the heating coils?
5. A 9 V battery maintains a current of 3 A through a radio. What is the resistance in the circuit?
6. If the voltage across a circuit is increased four times, what would you expect to happen to the current through the circuit?

7. (a) In the circuit below, calculate the resistance value of the resistor.

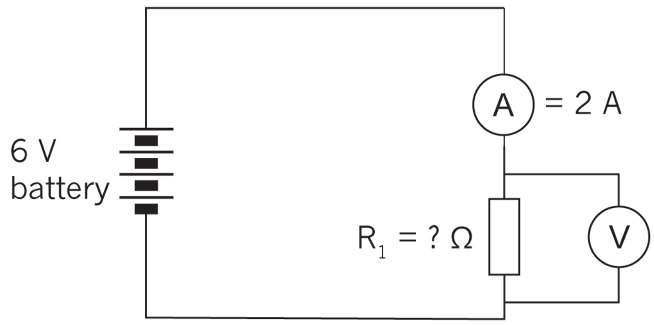



*Figure 16*

(b) If two more cells are added to the circuit, will the current increase or decrease? Check your prediction using the formula.

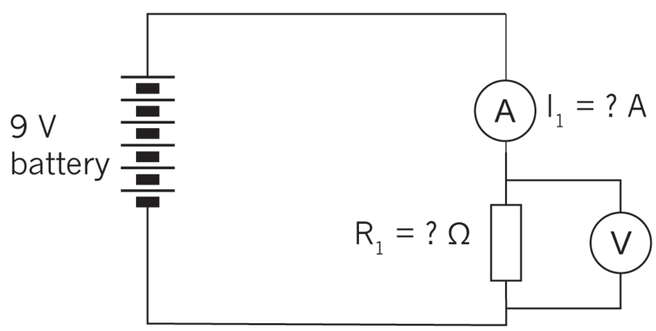



*Figure 17*

8. Calculate the battery voltage for the circuit below:

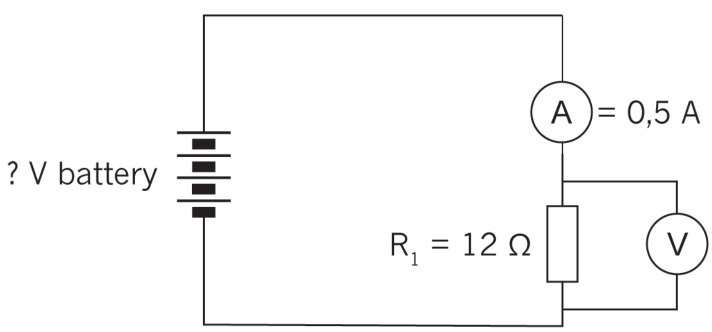



*Figure 18*

9. Look at the circuit below:

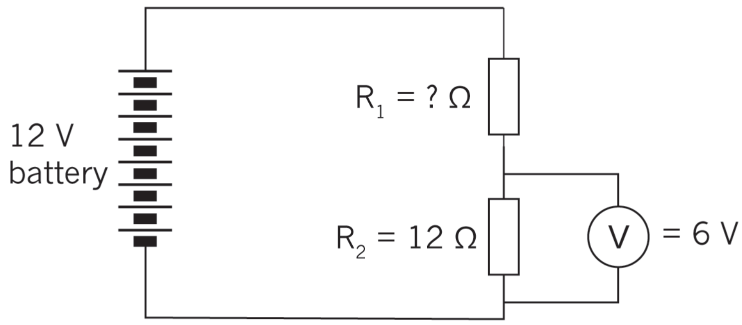



*Figure 19*

(a) Calculate the current through $R_2$.
(b) What will the current be through $R_1$?
(c) What will the voltage across $R_1$ be?
(d) What will the resistance value of $R_1$ be?

## Next week

In the next chapter, you will learn about components commonly used in electronic systems and their special functions.

# Chapter 13
# Electronic components 1

In this chapter, you will learn about electronic systems and about components in electronic circuits. You will also learn about the following control devices: switches, diodes and transistors. Finally, you will make a simple transistor circuit.



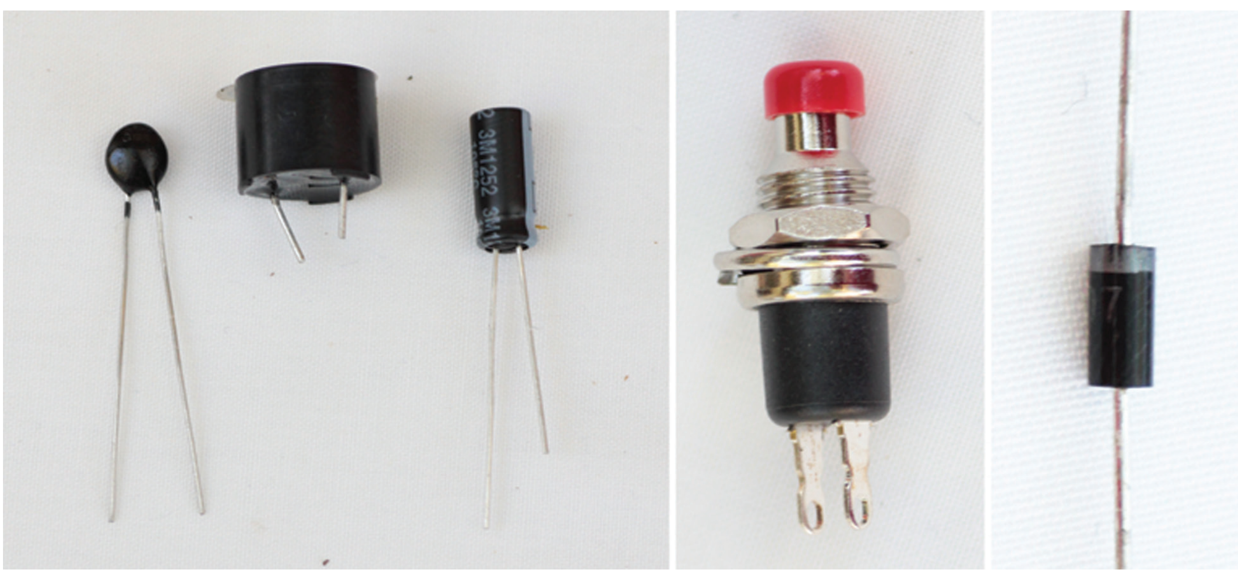


*Figure 1: A few examples of electronic components that we will deal with in this chapter*

## 13.1 Switches

A switch controls the electric current by closing or opening the circuit. There are various types of switches that control the circuit in different ways. In this lesson, you will learn about manual switches that a user can turn on or off.

### List different switches

1. Think about different switches that you use daily and make a list of as many of them as you can.

### Push button switch

Push button switches are often used for doorbell circuits, as in Figure 2. This simple doorbell circuit consists of cells in series, a push button and a buzzer, all connected by conducting wire. A person visiting the house presses the button for a short time and then releases it.

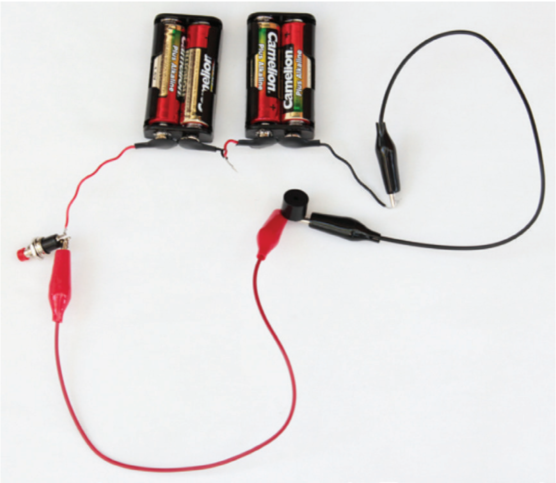


*Figure 2: A simple doorbell circuit*

### A doorbell circuit

1. Draw the circuit diagram of the doorbell circuit in Figure 2. Use the correct circuit diagram symbols. Note that the cells are in series.
2. Explain in your own words how this circuit works.

## Single-pole, single-throw switch (SPST)

Switches are named using the words "pole" and "throw". Pole refers to the number of circuits the switch controls, and throw refers to how many contacts the switch can make.

Single-pole, single-throw switches (SPST) control one input circuit and make one contact with the output circuit.

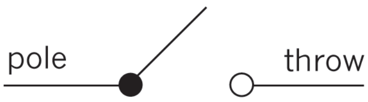



*Figure 3: The symbol for an SPST switch*

An example of an SPST is a light switch. Below is a typical lighting circuit.

When the switch is closed, the current will flow from the positive terminal (+) of the battery through the switch, through the lamp and back to the negative (–) terminal of the battery.

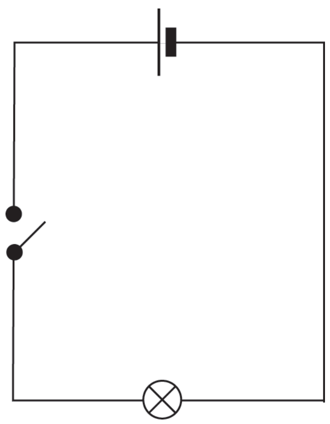


*Figure 4: A typical light circuit with an energy source, switch and lamp*

## Single-pole, double-throw switches (SPDT)

Single-pole, double-throw switches control one circuit, but they make two contacts so that they can control two devices. They turn on device 1 in one position and device 2 in the other position. There is no "off" position for this switch.

An example of an SPDT is a switch that turns on a red lamp in one position and a green lamp in the other position.

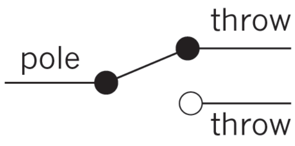



*Figure 5: The symbol for an SPDT switch*

## A lighting circuit

The circuit diagram below shows a two-way lighting circuit.

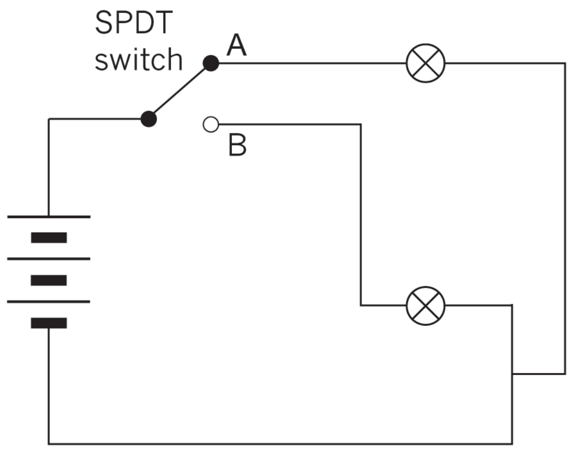



*Figure 6: A circuit with a battery, two lamps and an SPDT switch controlling two outputs*

1. Explain in your own words how this circuit works.
2. Think about how you can use an SPDT switch. You can make up an example, as long as it makes sense.
3. Look at Figure 6 again. An SPDT switch controls two possible outputs. They cannot both be ON, nor can they both be OFF. Is this an example of OR logic or AND logic? Explain your answer.
4. Look at the circuit diagram below. It shows how one light can be controlled by two different switches.

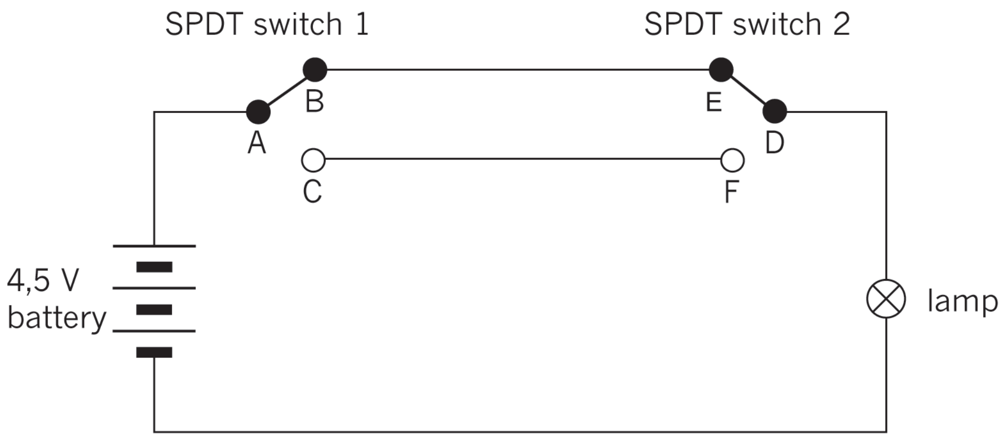



*Figure 7: A circuit with two SPDT switches is often used to control a lamp with one switch at each end of a long passage. It is also used to control a lamp with one switch at the bottom of a staircase, and the other switch at the top of the staircase.*

(a) Will the lamp turn on if A connects to C and D connects to F?

(b) Will the lamp turn on if A connects to C and D connects to E?

(c) Will the lamp turn on if AB and ED are closed?
(d) Will the lamp turn on if DF and AB are closed?
(e) Explain why the type of circuit in Figure 7 is useful for controlling the lamp in a long passage-way.

## Double-pole, double-throw switches (DPDT)

A double-pole, double-throw switch (DPDT) is like two SPDT switches with their switch levers attached to each other. There are two input circuits, and for each input circuit, there are two possible output circuits.

In the symbol below, the dotted line shows that the switches operate together.

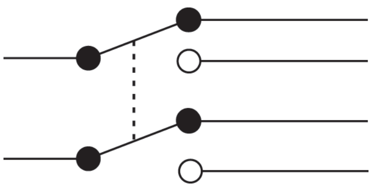


*Figure 8*

Consider an automatic car gate powered by an electric motor. To open the gate, the motor should turn in one direction. To close the gate, the motor should turn in the *opposite* direction. How can the direction in which the motor turns be changed? The way to do this is to change the direction of the current through the electric motor. Double-pole, double-throw switches can be used to reverse the direction of current through a circuit, so they are useful in applications such as automatic car gates. The circuit diagram below shows how a DPDT switch can change the direction of current through an electric motor.

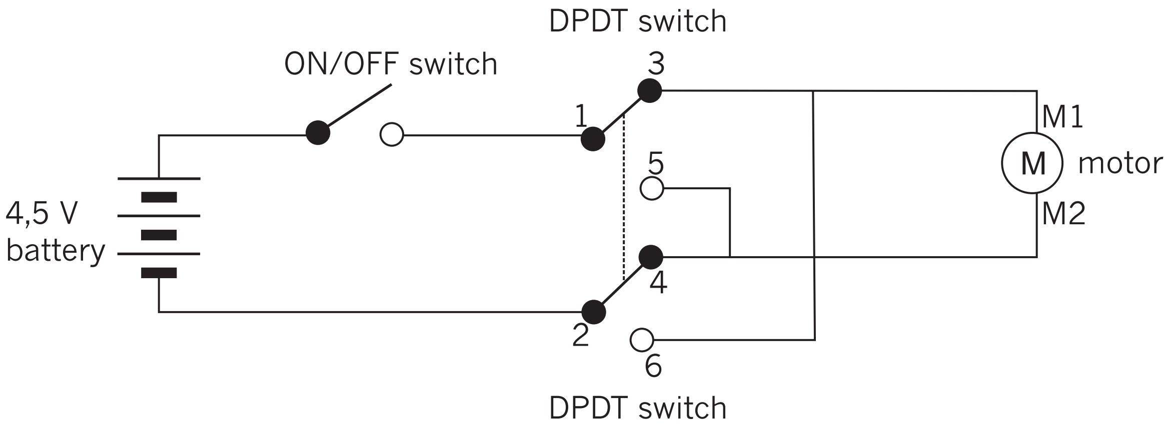



*Figure 9: A circuit where a DPDT switch controls the direction in which an electric motor turns*

The motor shaft will rotate in one direction when the current passes through it from terminal M1 to M2, but the motor shaft will rotate in the *opposite* direction when the current passes through it from terminal M2 to M1.

When the ON/OFF switch is switched ON, with the DPDT switch in the position indicated in the diagram above, the current will flow from the positive of the battery, through the ON/OFF switch to 1, to 3, through the motor from M1, to M2, to 4, to 2 and back to the negative of the battery.

When the DPDT switch is moved to the other position than in Figure 9, the current will flow through the circuit in the following order:

- from the positive terminal of the battery,
- through the ON/OFF switch to 1,
- through the top part of the DPDT switch from 1 to 5,
- through the motor from M2 to M1,
- to 6,
- through the bottom part of the DPDT switch from 6 to 2, and
- to the negative terminal of the battery.

## A gate motor circuit

1. Explain in your own words how this circuit works.
2. Explain the difference between an SPDT and a DPDT switch.

## 13.2 Diodes

A diode is a component with two terminals that can be connected in a circuit. The function of a diode in a circuit is to allow an electric current to flow in the forward direction and to block current in the opposite direction.

If the anode is connected to a higher voltage than the cathode, the current will flow from the anode to the cathode. This is called “forward bias”.

If the diode is put in the circuit back to front, so that the voltage at the cathode is higher than the voltage at the anode, the diode will not conduct electricity. This is called “reverse bias”.

Diodes are normally used to prevent damage to other components in circuits. For example, some components have positive and negative terminals and will be damaged if a current goes through them in the wrong direction. A diode can protect against a current flowing the wrong way if a battery was put in incorrectly to power the components. If you put batteries into a radio incorrectly, a diode will prevent damage to the radio.

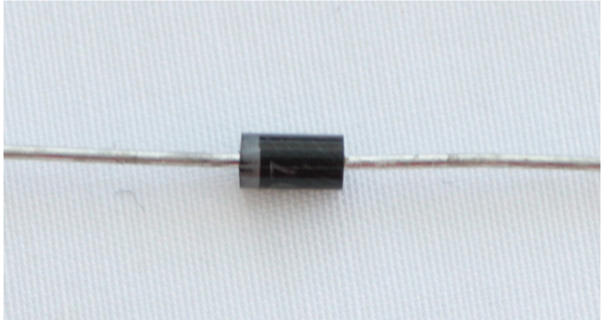


*Figure 10: A diode*

Diodes vary considerably in size, current-carrying capacity, and reverse blocking voltage. They range from small diodes that can only handle 20 mA with a reverse blockage of 30 V, to large industrial diodes that can carry hundreds of amps and block up to thousands of volts. You can use a multimeter or a simple tester (battery, resistor and LED) to check in which direction a diode conducts.

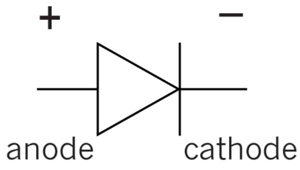



*Figure 11: Circuit symbol of a diode. The current flow in a diode is in the direction of the arrow head.*

## Light-emitting diodes (LED)

A light-emitting diode (LED) is a special kind of diode that glows when electricity passes through it. LEDs produce light of specific colours, based on the materials they are made from. For example, they can produce red, amber, yellow, green, blue, violet and white. The most common colour is red.

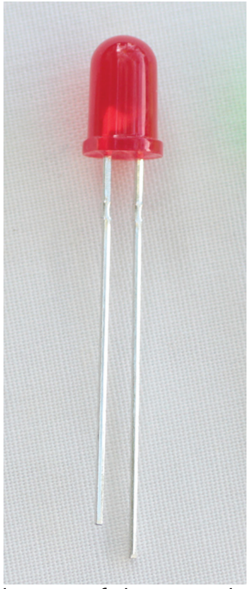


*Figure 12: An LED. The longer of the two wires coming out of the LED should be connected to the positive terminal, and the shorter wire to the negative terminal.*

LEDs are often used to show if a circuit is working. Think about the small red light glowing on the front of a TV set that can sometimes change from red to amber.

LEDs are used as indicators in many devices, including calculator screens and digital clocks.

The LED will only allow current to pass in one direction. The cathode is normally indicated by a flat side on the casing and the anode is normally indicated by a slightly longer leg. The current required to power an LED is usually around 20 mA.

The arrow symbol for an LED tells you in which direction the current flows.

Nowadays, LEDs are used in many cases where normal light bulbs were used. For example, household lighting is being replaced by LEDs. They are replacing light bulbs because they are more efficient and use much less electric energy. They also last for a long time.

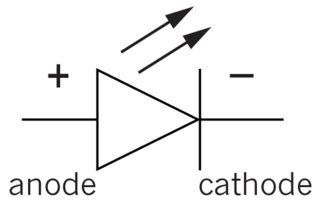



*Figure 13: The circuit symbol for an LED.*

To protect an LED from too much current, a resistor has to be added to the circuit, as in the diagram below.

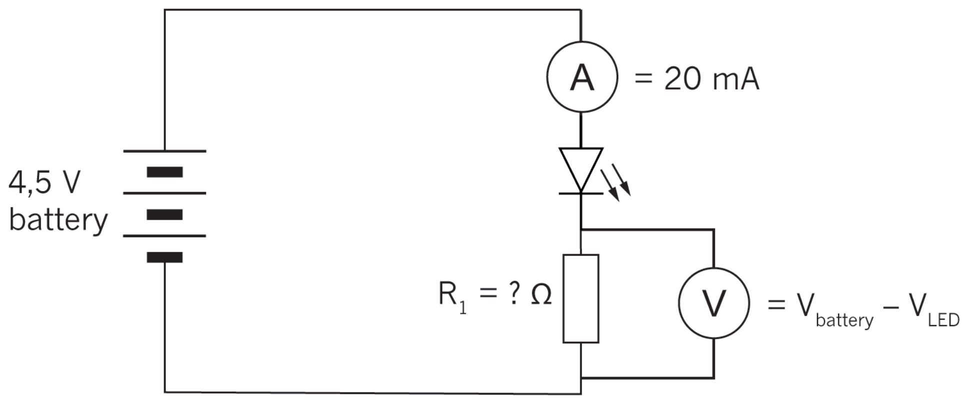



*Figure 14: LED circuit with a current-limiting resistor.*

## Questions

1. Describe the function of a diode in your own words.
2. List at least four places where LEDs are used. Don't use the examples already given.
3. How can you make sure that a diode is connected correctly?
4. Draw the circuit symbols for a diode and for an LED.

## 13.3 Transistors

Transistors are very important building blocks of modern electronic devices. They enable us to design smaller and cheaper electronic devices.

A transistor is a semiconductor device that consists of three layers. Each layer has its own connection point with a specific name: **collector**, **base** and **emitter**.

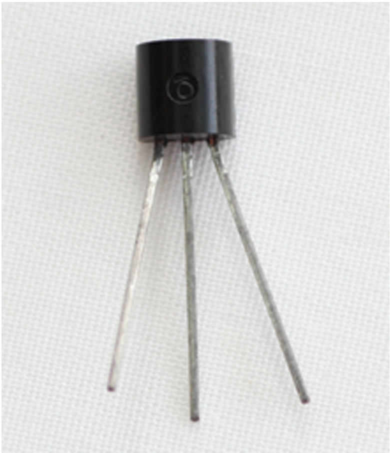


*Figure 15: One type of transistor*

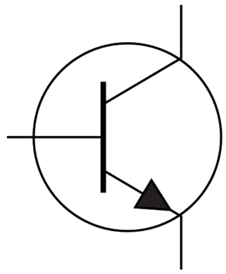



*Figure 16: The circuit symbol for an npn transistor*

A transistor works as a type of switch to turn current on and off. It can also amplify a current.

An **npn** transistor acts as if there is a switch between the collector and the emitter. When the voltage drop between the base and the emitter is smaller than 0,6 V, the resistance between the collector and the emitter is very high, so only a very small current passes through. When the voltage drop between the base and the emitter goes higher than 0,6 V, the resistance between collector and emitter suddenly decreases by a large amount, and then a strong current flows through the transistor. The transistor works like a switch because a tiny current through the base, going to the emitter, switches on a much larger current through the collector. So it is an electrically controlled switch.

Transistor is short for "trans-resistor" and this explains how it works. With a relatively small base current, the resistance between the collector and the emitter is changed. As the base current increases, the collector emitter resistance decreases.

In Chapter 14, you will learn about the applications of transistors.

There are other types of transistors, for example pnp transistors that work a bit differently from npn transistors. But you will only work with npn transistors in this term.

## A transistor circuit

Suppose you want to make a switch that is ON or closed when you touch its two terminals with your finger, and OFF or open when you don't touch it. Look at the circuit diagram in Figure 17 for a touch switch such as the one described. The purpose of this circuit is to light up the LED when you touch the touch switch with your finger.

Unfortunately, this circuit won't work well, since your finger is a very weak conductor. In other words, it has a very high resistance. So the current will be very small when you touch the switch. Therefore, the LED will only emit a dim light.

By using a transistor, you can build a circuit that uses the very small current through your finger to switch on a larger current that passes through the LED, which will then emit a bright light.

Figure 18 shows a circuit that uses a transistor for this purpose. In this circuit, the touch switch is an "input device," the *npn* transistor is a "control device," and the LED is the "output device".

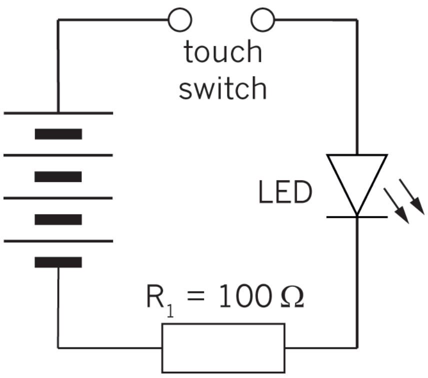



*Figure 17: A simple touch switch circuit that will not work well*

A transistor uses a small current circuit to switch on a larger current circuit. This is why transistors are also used in music equipment to "amplify" the sound.

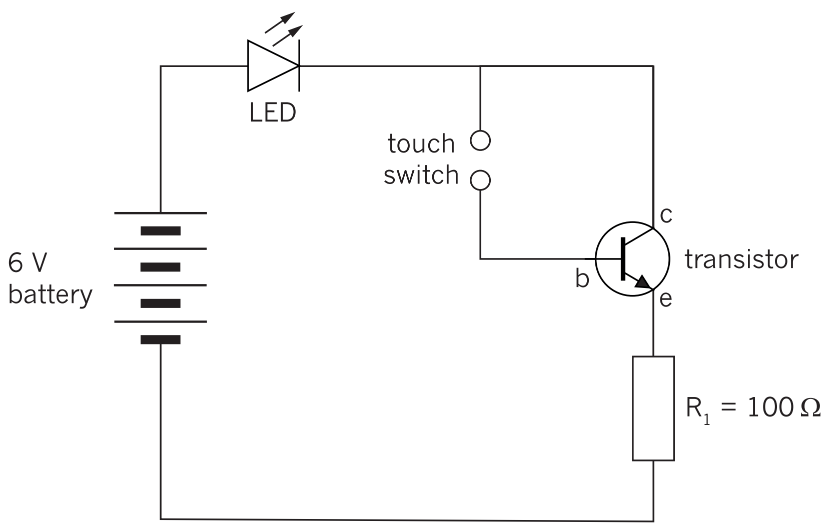



*Figure 18: Circuit using a transistor as an electronic switch*

1. The photograph below shows a circuit built according to the circuit diagram in Figure 18. Look at the photograph and identify each component in the circuit. Redraw the photograph and write labels for the different components and draw arrows pointing from the labels to the components.

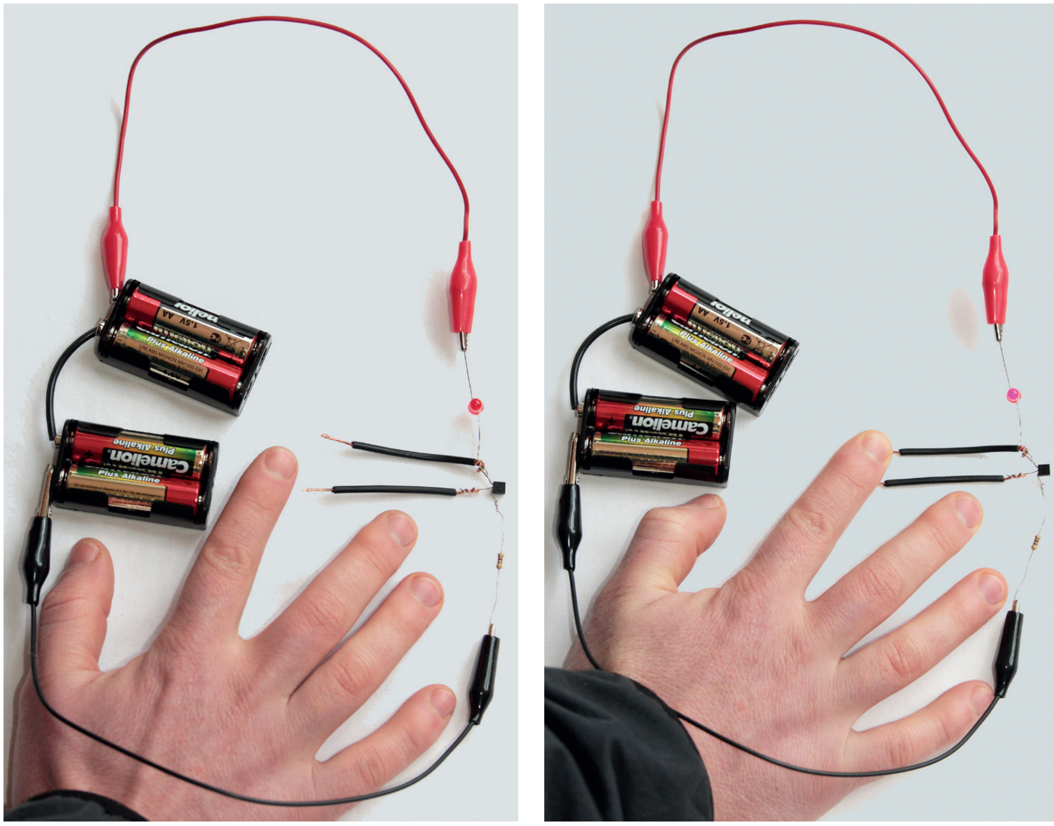


*Figure 19: The construction of a touch switch circuit with a transistor and an LED.*

2. Explain how the different parts of the transistor are connected in this circuit.
3. Explain what you expect to see when the touch switch is activated.
4. Touch the two terminals of the touch switch with one finger. Describe what happens.

## Next week

Next week, you will learn more about electronic systems and components in electronic circuits. You will also learn about capacitors, and various kinds of input devices such as sensors.

# Chapter 14
# Electronic components 2

In this chapter, you will learn more about electronic systems and components in electronic circuits. You will learn about various kinds of sensors that act as input devices. A touch switch is a sensor that works with the moisture on your skin. This is a very sensitive device that produces a small current. A transistor is required to make the current big enough to have an effect. This week, you will learn about other kinds of sensors and how they are used in devices.

You will also learn about capacitors.



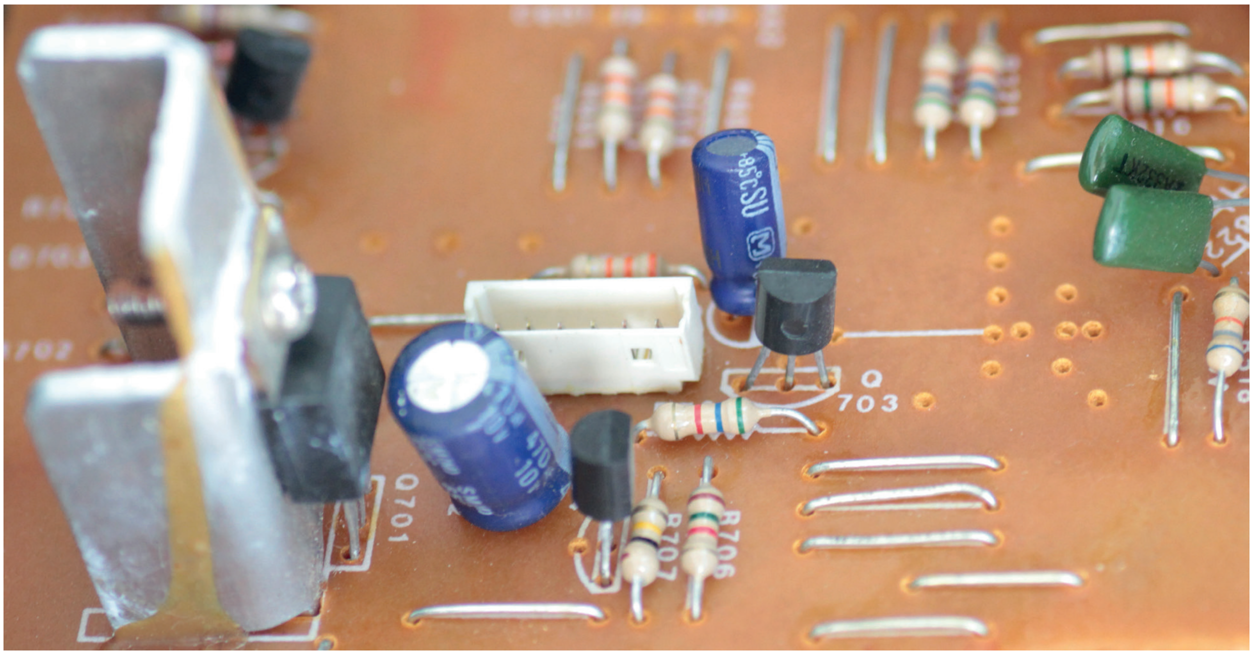


*Figure 1: Components connected on a printed circuit board*


*Figure 2: Measuring the resistance of a thermistor at room temperature.*

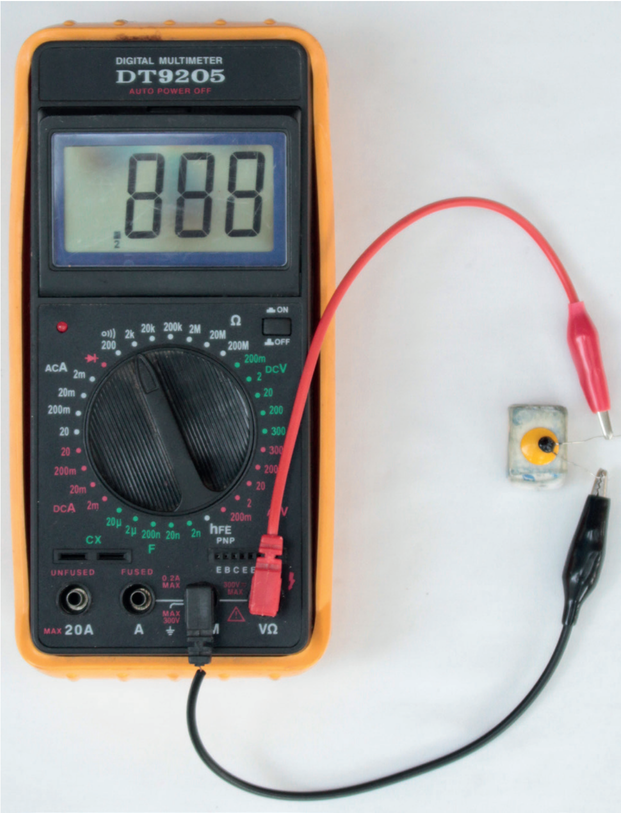



*Figure 3: Measuring the resistance of a thermistor while heating it with a hot object. You can heat a metal thumb tack by pressing it into an eraser, and then rubbing it hard against a piece of wood or plastic for one minute.*

***Safety warning:*** *The thumb tack can get very hot and burn your skin, which can cause a wound.*

## 14.1 Light-dependent resistors (LDR)

A light-dependent resistor, also called an **LDR**, is a resistor of which the resistance *decreases* when it is exposed to light of a higher intensity. It can therefore be used to detect light and trigger warning devices in cases where light may cause problems.

The resistance of an LDR increases when it becomes darker.

- When an LDR is in the dark, its resistance value will be very high, around 1 MΩ.
- When an LDR is exposed to a light of high intensity, the resistance value will decrease. It could drop from 1 MΩ to 2 kΩ.

An LDR has two terminals that can be connected to a circuit in either direction.

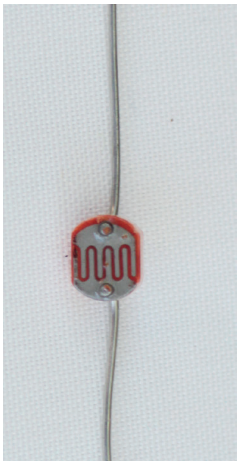


*Figure 4: A light-dependent resistor*

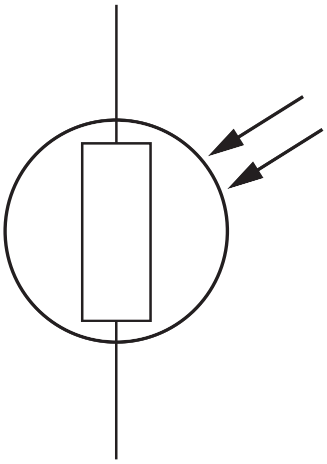


*Figure 5: The circuit symbol for a light-dependent resistor*

## Circuit of a day/night switch

Day/night switches are often used to turn on street and outside lights once it gets dark. It has an advantage above time switches, since the time settings can go wrong, and the amount of daylight does not remain constant during different weather conditions.

In this example, a light-dependent resistor (LDR) is the input device, an *npn* transistor is the control device, and an LED is the output device.

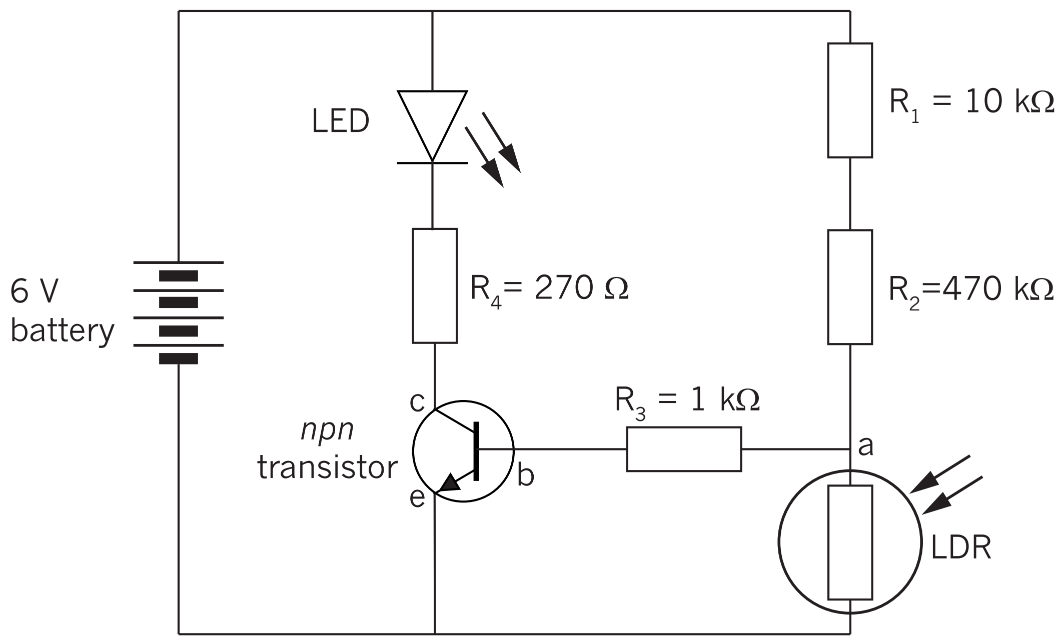



*Figure 6: Circuit diagram of a day/night switch*

1. Write four examples of when it would be useful to have a device that detects the amount of light, and does something in response to it.
2. What is the role of the LDR in the circuit?
3. Describe how the transistor is connected to the circuit.
4. What is the role of the transistor in this circuit?

## 14.2 Thermistors (temperature-sensitive resistors)

The resistance value of this resistor depends on the temperature it is exposed to. There are two types of thermistors:

- A "negative-temperature coefficient" type thermistor, where the resistance value decreases with an increase in temperature.
  This is also called an "NTC" or "–T" thermistor.
- A "positive-temperature coefficient" type thermistor, where the resistance value increases with an increase in temperature.
  This is also called a "PTC" or "+T" thermistor.

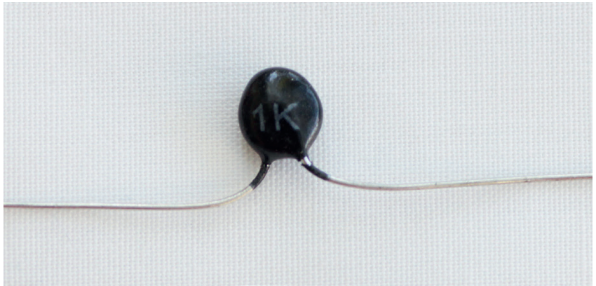



*Figure 7: A thermistor*

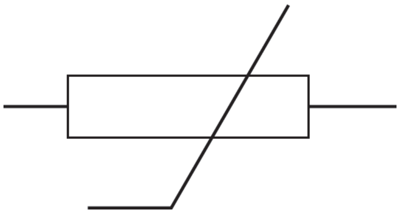


*Figure 8: The circuit symbol for a thermistor*

### Uses of thermistors

1. Write four examples of situations in which electronic devices that use a thermistor of either type would be useful.

### Measuring the resistance of a thermistor

Figures 2 and 3 on page 176 show the resistance of a thermistor measured at room temperature, and when heated by placing it on a hot object. At room temperature, the resistance is 1 413 Ω. When the thermistor is heated with a hot object, the resistance decreases to 888 Ω.

1. Was the thermistor a PTC or an NTC?
2. Give reasons for your answer.

## Heat-activated switch

A thermistor can be used in a heat-controlled switch for a fire alarm. When the thermistor is heated up, its resistance is decreased and the transistor starts conducting a current, switching on the LED.

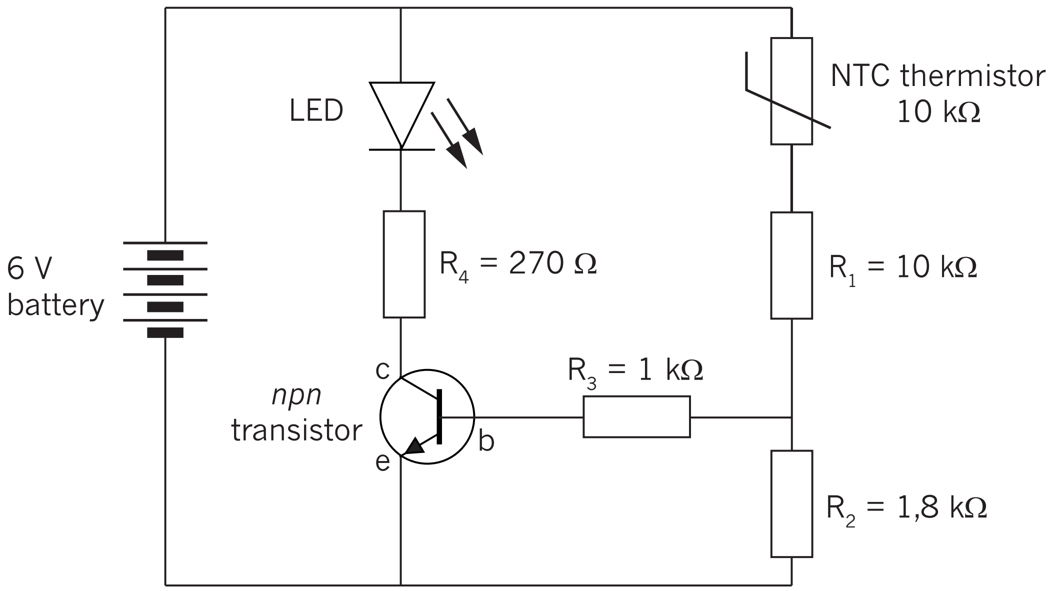



*Figure 9: Diagram of a simple fire alarm with an NTC thermistor*

1. What is the role of the thermistor in the circuit?
2. Describe how the transistor is connected to the circuit.
3. Draw a simplified circuit diagram for an indicator light to show when a heater has dropped below a certain temperature and starts heating up again.

## 14.3 Capacitors

A capacitor stores the potential energy that results from separating positive and negative charges. A capacitor consists of two metal plates separated by an insulator called a dielectric. The ability of a capacitor to separate electric charge is called its capacitance.

Capacitance is measured in farad. The symbol "C" is used for capacitance. Because the farad is such a large unit, practical values usually have the prefixes m (milli-), µ (micro-), n (nano-) or p (pico-).

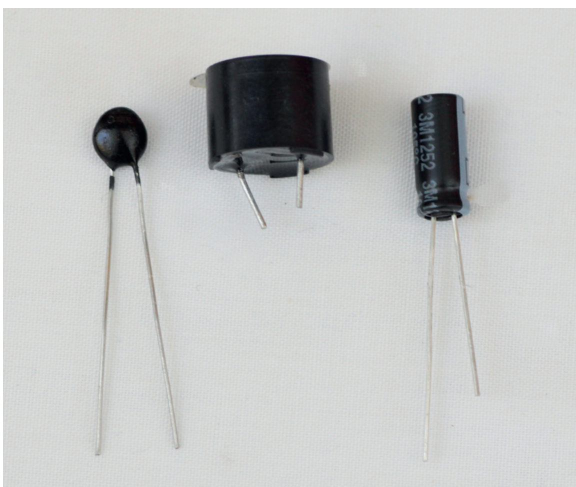


*Figure 10: Different types of capacitors*

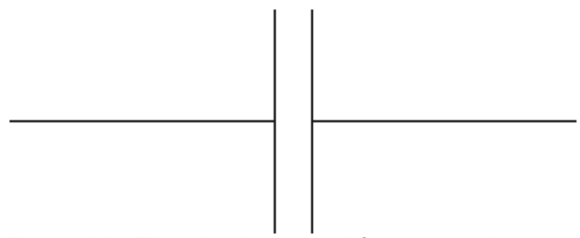


*Figure 11: The circuit symbol for a capacitor*

When capacitors are connected in parallel, the total area of the metal plates on each side is increased, so the total capacitance is increased.

When capacitors are connected in series, the distance between the opposite plates is increased. And because capacitance is inversely proportional to the distance between the plates, the total capacitance is reduced to less than that of the smallest capacitor.

### Charge and discharge of a capacitor

The charging and discharging of a capacitor can be observed by building the circuit in the diagram below. When the switch is switched to position A, the current will flow from the + of the battery, through $LED_1$, through the switch to one plate of the capacitor. The negative of the battery is connected to the other plate of the capacitor through the resistor $R_1$. While the capacitor is charging, $LED_1$ will be ON.

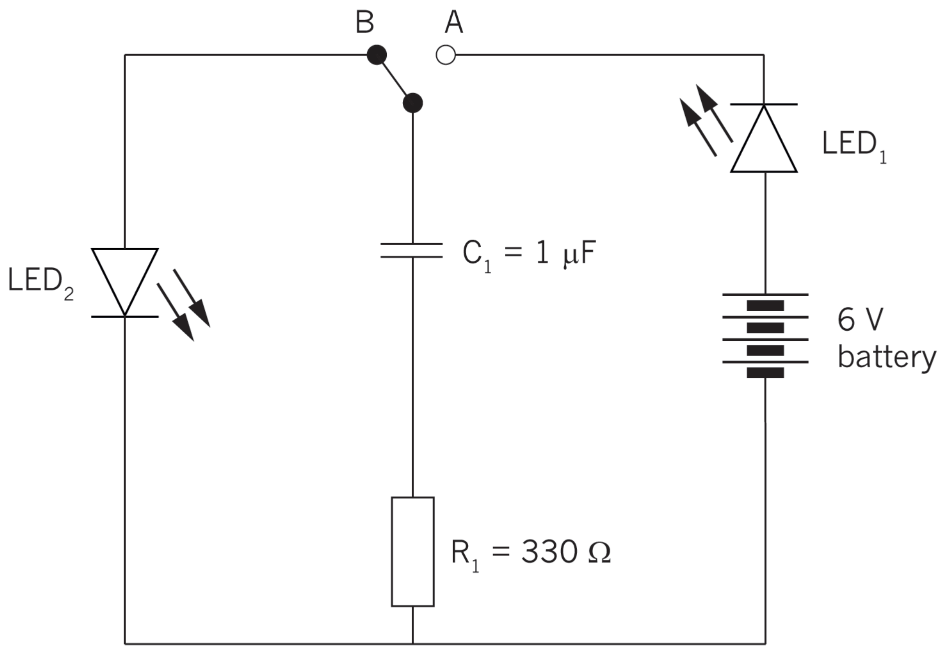


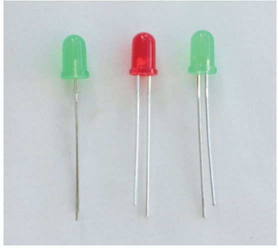
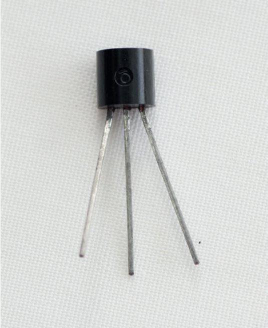
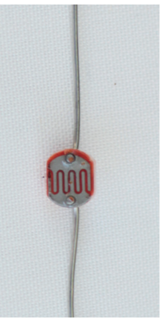

*Figure 12: Capacitor charging and discharging circuit*

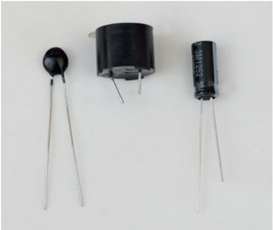
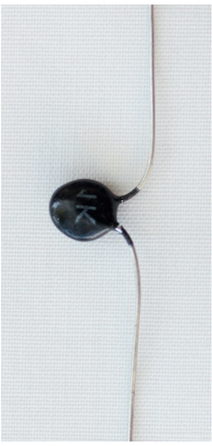
After the capacitor has been charged and the switch is switched to position B, a current will now flow from the + plate of the capacitor through $LED_2$, and through the resistor $R_1$ to the negative plate of the capacitor. While the capacitor is discharging, $LED_2$ will be ON.

Capacitors are often used in electronic devices that need a carefully controlled time delay, such as timers and traffic lights. The exact kind of capacitor can be chosen to get the exact time delay that is needed. Increasing the value of the capacitor increases the length of the time delay.

## Questions about components

1. Copy the table on the next two pages, without the photographs. Name the component in the picture and draw the correct circuit symbol next to the name. Write a brief description of the main uses of the component.

| Name of component | Picture | Symbol | Use |
|---|---|---|---|
| |  | | |
| |  | | |
| |  | | |

| Name of component | Picture | Symbol | Use |
|---|---|---|---|
| |  | | |
| |  | | |
| |  | | |

## Next week

Next week, you will draw circuit diagrams and build simple circuits.

# Chapter 15
# Build and draw electronic circuits

In this chapter, you will draw circuit diagrams and assemble four electronic circuits, using the components you have learnt about in Chapters 13 and 14.



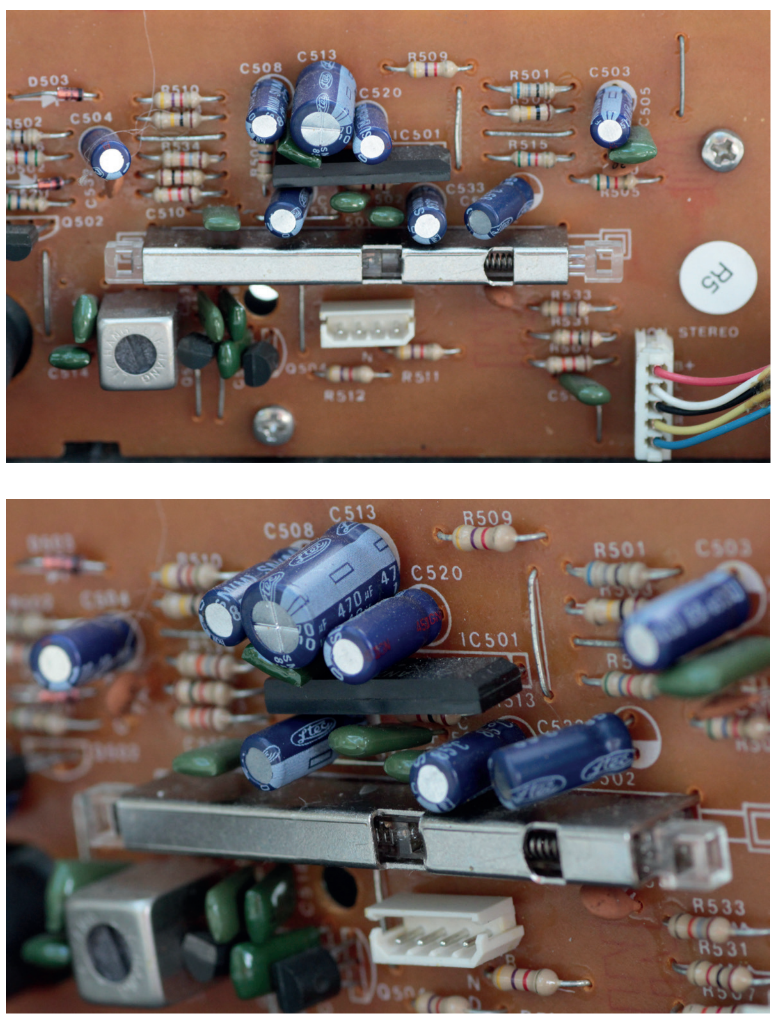



*Figure 1: A part of the circuit for a radio*

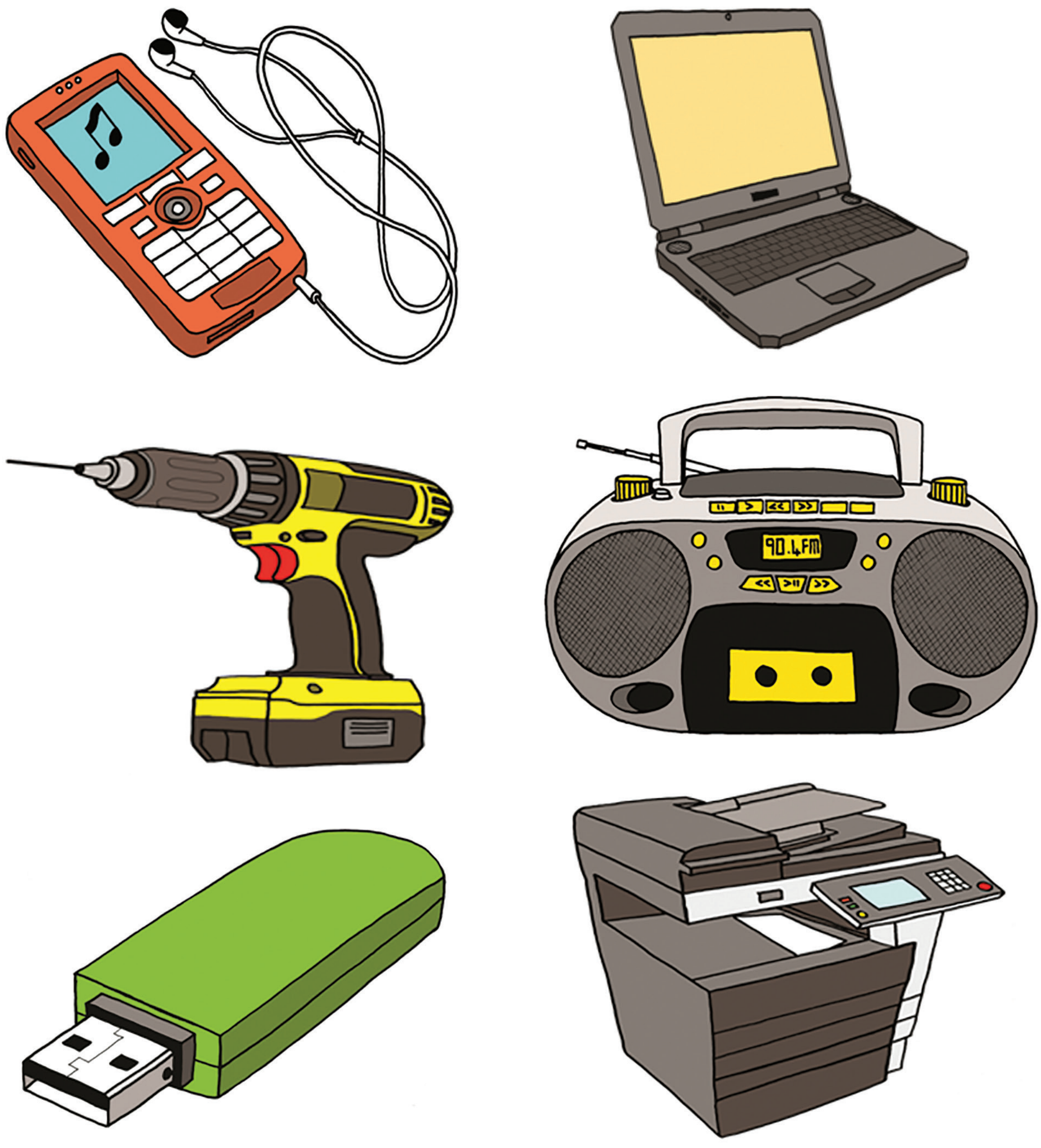



*Figure 2: All of these appliances contain electronic circuits.*

# 15.1 Simple electronic circuits

## A circuit with an LED

In this lesson, you need to assemble a simple LED circuit. You will draw the circuit diagram on your own and then work in pairs to assemble it.

**You will need:**

- an LED,
- a 470 Ω resistor,
- a switch,
- four 1,5 V cells in series, or a 9 V battery, and
- electric conducting wire with crocodile clips for connections.

The photograph below shows the circuit you need to build.

1. Draw a circuit diagram for Figure 3.

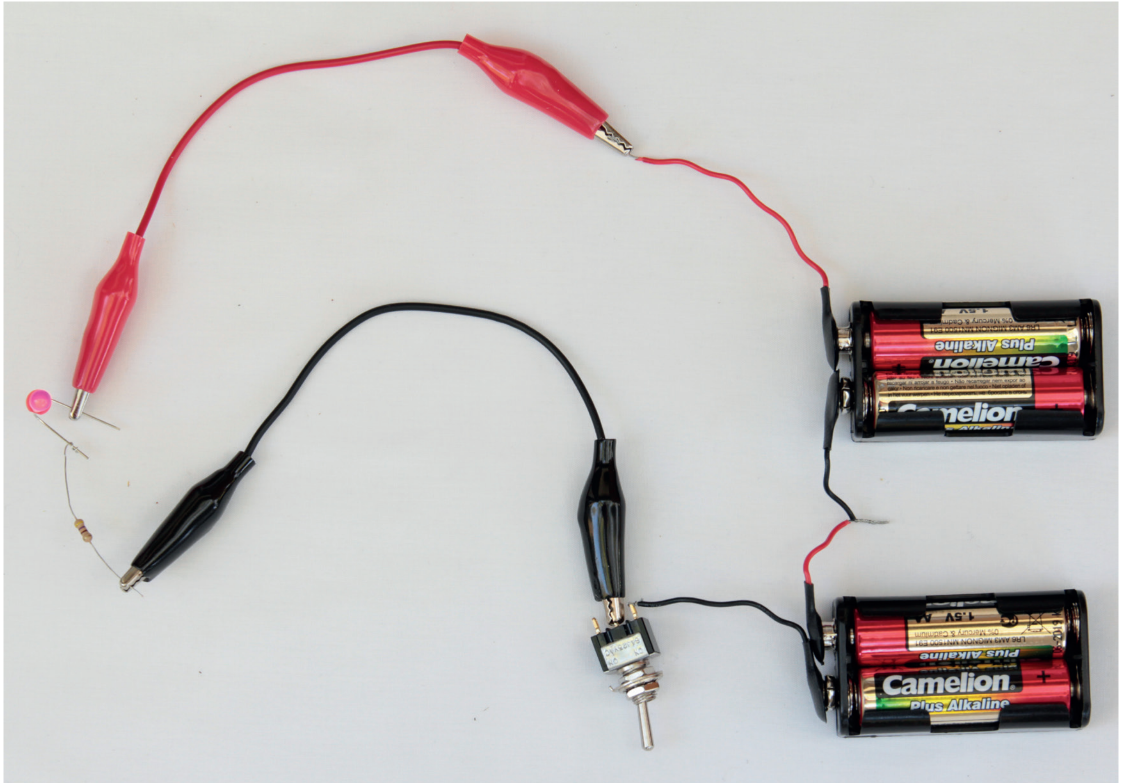



*Figure 3: A circuit with an LED, a battery, a switch and a resistor*

## A circuit with an LDR

Now you will build a circuit where an LDR regulates the current.

**You will need:**

- an LDR,
- four 1,5 V cells in a cell holder, and
- a buzzer.

The photograph below shows a circuit in which an LDR regulates the current through the circuit.

*Figure 4: A circuit where the current is regulated by a light-dependent resistor*

1. Work individually to draw a circuit diagram of Figure 4.
2. Work in pairs to build the circuit.
3. Predict what will happen when:
   (a) The LDR is covered
   (b) The LDR is in bright sunlight
4. Is the buzzer affected by different sources of light, such as light from an electric lamp, light from a cell phone screen or light from a torch? Do a practical investigation and write down your findings.

## 15.2 A control circuit and a time-delay circuit

### A fire alarm: A circuit with a sensor and a transistor

In the next lesson, you will build the electronic circuit for a fire alarm. In the next chapter, you will use the same circuit but for a different purpose, as part of an automatic kettle switch. It is very important that you complete the circuit and that it works, as you will use it in the PAT in the weeks that follow.

The type of circuit you will build is used very often to switch an *output device* on and off without using a switch. Instead of a switch controlled by hand, this type of circuit uses an *input sensor* in combination with a transistor to switch the output device on or off *automatically*, depending on the measurement of something by the input sensor.

This type of circuit is called a *control circuit* since one circuit controls another circuit. In the case where a transistor is used with a sensor such as an LDR, the base-emitter current controls the larger collector-emitter current.

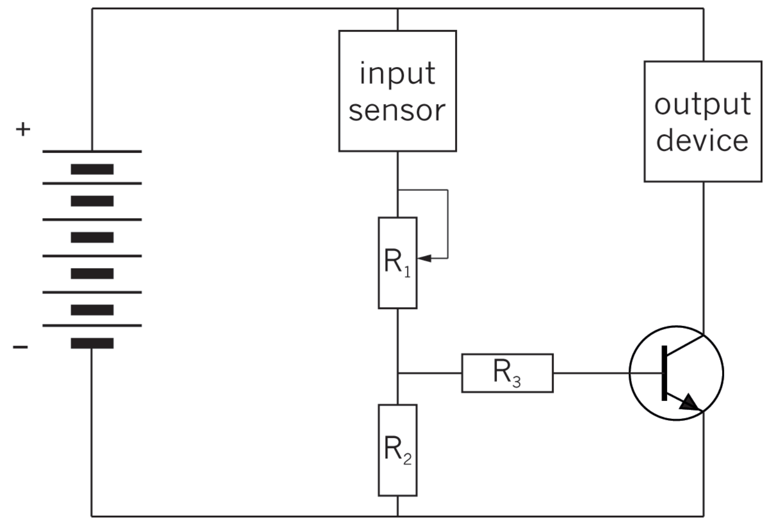



*Figure 5: The circuit diagram for the control circuit*

Note that resistor 2 and the input sensor may have to change places depending on the relationship between the resistance of the input sensor and the required output:

- If a *decrease* in resistance of the input sensor should switch on the output device, then resistor 2 and the input sensor should be arranged as in Figure 5. Look back at the circuit for a heat-activated switch using a negative temperature coefficient (NTC) thermistor, on page 180.
- If an *increase* in resistance of the input sensor should switch on the output device, then resistor 2 and the input sensor should be arranged the other way around to Figure 5. Look back at the circuit for a day/night switch using a light-dependent resistor (LDR), on page 178.

It is easier to understand the circuit if you think about a systems diagram. Look at Figure 6. The yellow part is the output side of the diagram.

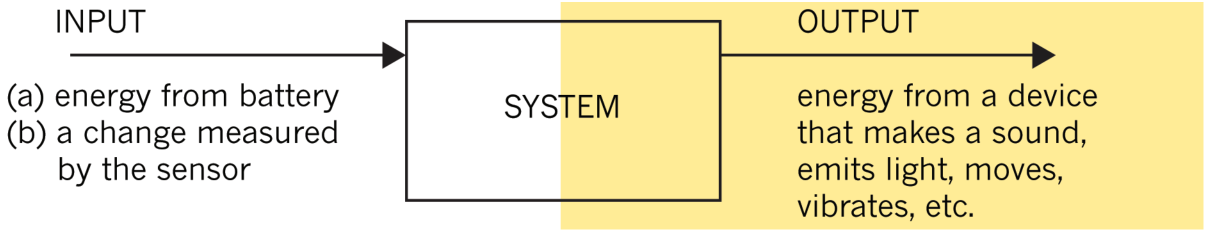



*Figure 6: A systems diagram of a control circuit*

Figure 7 shows how the circuit in Figure 5 is the same as the systems diagram.

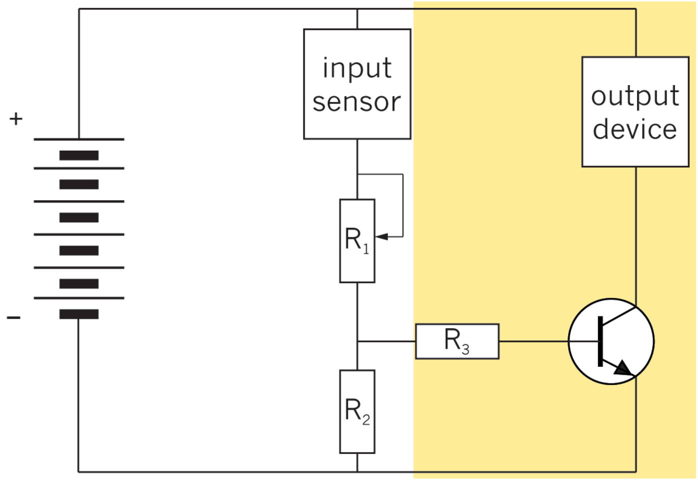



*Figure 7*

## Identify the different components and draw the circuit

The circuit for the fire alarm contains the following components:

- a battery consisting of 4 cells in series,
- an input sensor to measure the temperature,
- a variable resistor to set the temperature at which the alarm should go off,
- an output device to make noise when it gets too hot, and
- a transistor to switch the output device on when it gets too hot.

1. What type of electronic component will you use as the input sensor?
2. What type of device will you use as the output device?
3. What voltage does the battery supply to the circuit?
4. Draw a circuit diagram for a fire alarm:
   (a) Show the correct symbols for the components you will use as the input and the output sensors.
   (b) Show the voltage of the battery.
   (c) Show the emitter ("e"), base ("b") and collector ("c") of the transistor. Look back on what you learnt about transistors in Chapter 13.

The purpose of resistors 1 to 3 in the control circuit is hard to explain. It has to do with the minimum current to the base of the transistor that is needed to allow current through from the collector to the emitter of the transistor. If you choose to study more electronics in FET or at university, you will learn about the purpose of these resistors, and how to calculate their resistances.

Someone has already done the calculations of the resistances of different components that should be used for the fire alarm to work. These are called the *specifications* for the resistances of components.

- $R_1$ = 700 to 1 400 kΩ (variable resistor)
- $R_2$ = 820 Ω
- $R_3$ = 1 kΩ
- input sensor: 10 kΩ

5. Show the specified resistances of the components on your circuit diagram.

## 15.3 Build a fire-alarm circuit

### Build a circuit and test it

Work in pairs to build the circuit.

**You need the following materials to build the circuit:**

- four 1,5 V cells in series, in a cell holder,
- conduction wires with crocodile clips,
- a 10 kΩ NTC thermistor,
- a 700 to 1 400 kΩ variable resistor,
- a 820 Ω and a 1 kΩ resistor,
- an *npn* transistor, and
- a buzzer, that is specified to be used with between 3 V and 6 V across it.

1. Now build the circuit. Set the variable resistor to its lowest resistance.
2. Once your circuit is complete, check that all your connections are good.
3. Then connect the battery to the circuit.
4. To test the fire alarm, warm up a thumb tack by pressing it into an eraser, and rubbing it hard against a piece of wood or plastic for a minute. Then press it against the thermistor.

**Troubleshooting**

- test whether the battery is flat or not,
- test all your connections again,
- follow the flow of the current on your board with your finger, to check whether you connected the components the right way, and
- check that you connected the transistor the right way round.

If the fire alarm makes a sound even when the thermistor is not heated, then increase the resistance of the variable resistor until the alarm stops making a sound. Do not increase the resistance of the variable resistor more than is necessary, because then the fire alarm will not make a sound when it is heated.

## If you have time: Build a time-delay circuit

Capacitors are often used in time-delay circuits.

**You will need:**

- four 1,5 V cells in series, or a 9 V battery,
- two LEDs,
- a 470 Ω resistor,
- a 1 000 μF capacitor, and
- an SPDT switch.

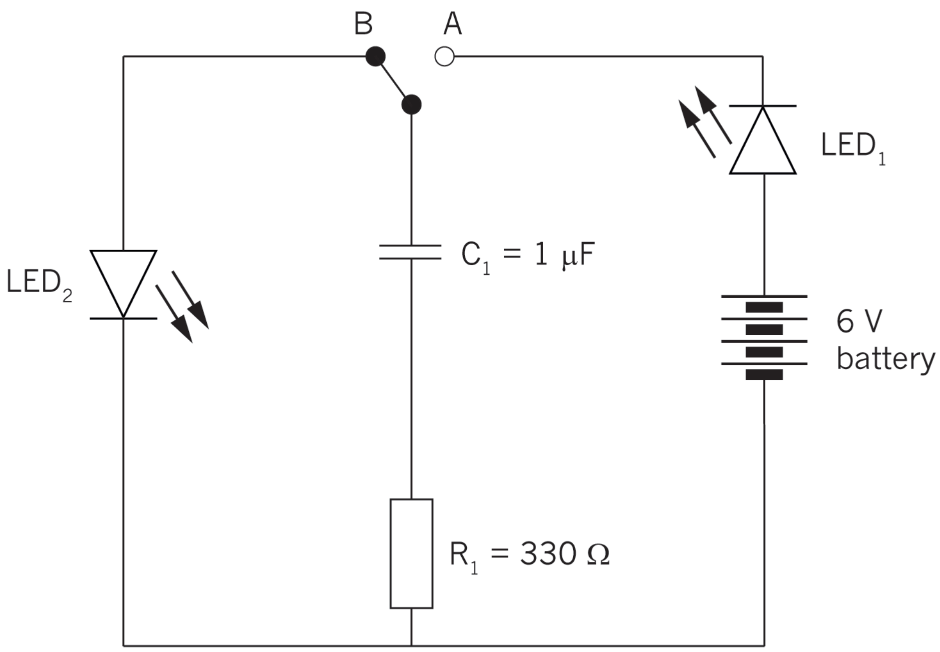



*Figure 8: A time-delay circuit*

1. Build the circuit. Put the switch to A and observe the LEDs. Describe what happens and explain it in detail.

## Further reading: Boards on which more complicated circuits are built

If you try to build a more complicated circuit by connecting components using conducting wire and crocodile clips, many wires will cross one another and the circuit will be messy, looking like a tangled bunch of ropes.

To make a complicated circuit in a neater and smaller way, most circuits are built on boards such as "bread boards", "strip boards", or "printed circuit boards" (PCBs).

Figure 9 below shows a simple LED circuit, such as the one you built in section 15.1, but here it is built on a strip board. Notice that there are no connecting wires used to build this circuit! This is because at the bottom of the strip board there are parallel copper strips connecting the holes in each column. This makes it possible to construct a circuit without using wire.

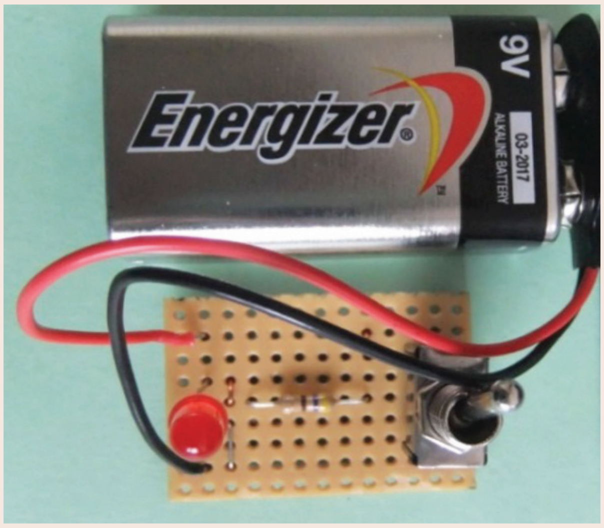



*Figure 9: A simple LED circuit built on a strip board*

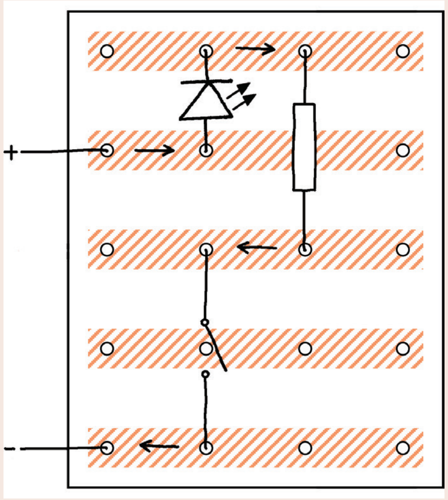



*Figure 10: One possible layout of the simple LED circuit on a strip board*

Figure 10 shows one possible plan of how to arrange the simple LED circuit on a strip board. The copper strips are at the bottom of a strip board, and not visible from the top. Therefore, the copper strips on the drawing of the layout were drawn with hatching, to show that you cannot really see them from the top.

The arrows on Figure 10 are drawn to help you understand how current flows through the copper strips at the back of the strip board. The current flows in the direction of the arrows.

The connectors of the components are soldered to the copper strips at the bottom of the strip board. This is to ensure that they make proper electrical contact with the copper strips.

Soldering is done with lead, because lead is a good electrical conductor and has a low melting point, so it is easy and quick to melt it with a soldering iron.

Bread boards and printed circuit boards are other types of boards used to build complicated circuits. They also have copper connections at the back, but these connections are arranged in a different way than on a strip board.

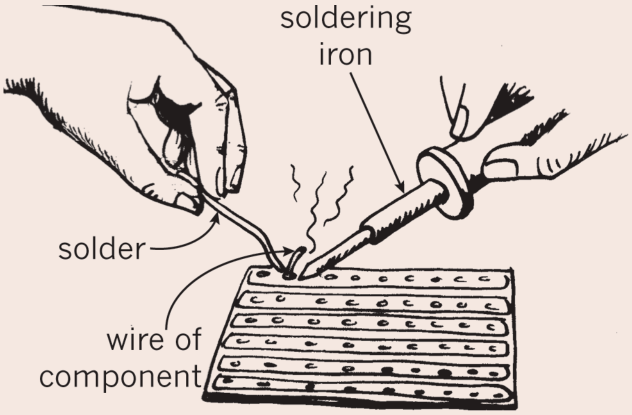



*Figure 11: Soldering components onto the back of a strip board*

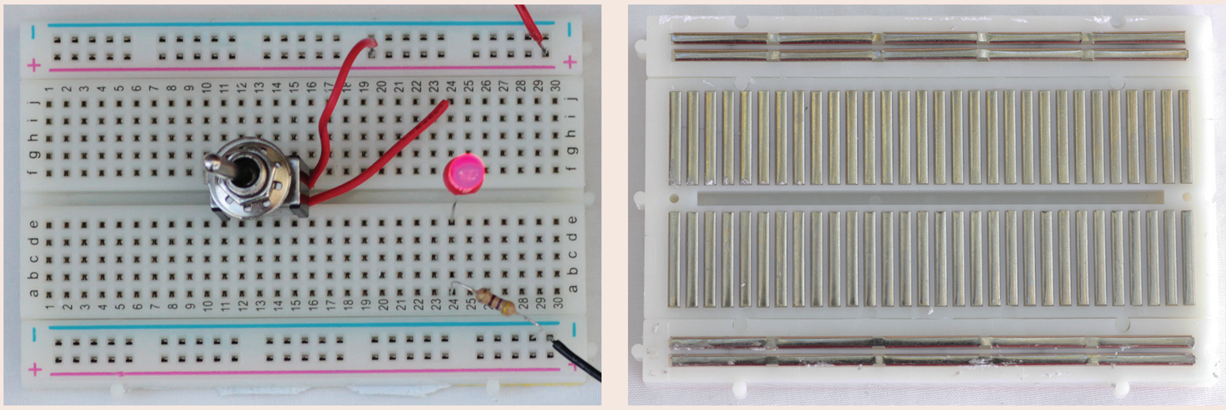


*Figure 12: The front and back of a bread board*

With a breadboard it is not necessary to solder connections, since each hole in the breadboard has a spring that grips the wire tightly to make proper electrical contact.

Almost all manufactured electronic devices use printed circuit boards, where the copper connections at the back can be made in any pattern. This makes it possible to make complicated circuits that are very small.

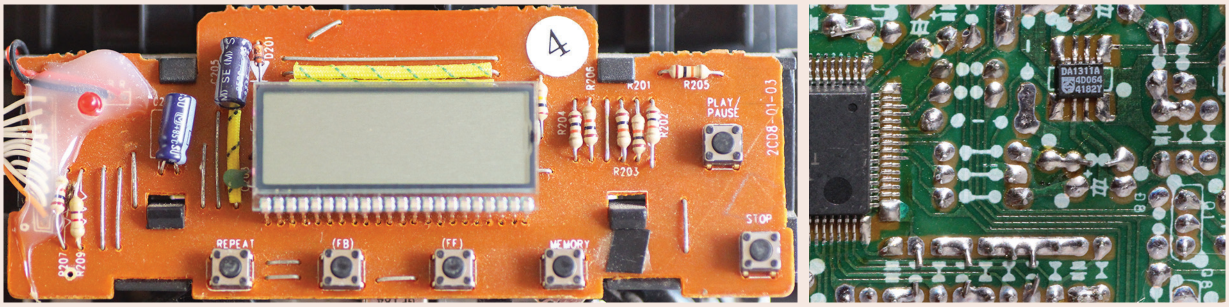


*Figure 13: The front and back of a printed circuit board*

## Next week

The next chapter is your PAT for this term. You will learn how an electronic circuit can be used to control another circuit with a much bigger current. You will build a device using both circuits and then test it.

# Chapter 16 PAT
# Electronic systems and control

In this PAT, you will first study where electronic circuits, using very small currents, are used to control electric circuits with much bigger currents. You will then design and build your own electric circuit that will be controlled by an electronic circuit.



**Assessment**

Situations where electronic systems control electric circuits (individual work) .......... [5]

Design brief and sketches (individual work) .......... [12]

Evaluate and improve the design (team work) .......... [8]

Final drawings of the design (individual work) .......... [15]

Make the switch (individual work) .......... [25]

Presentation (team work) .......... [5]

[Total: 70]

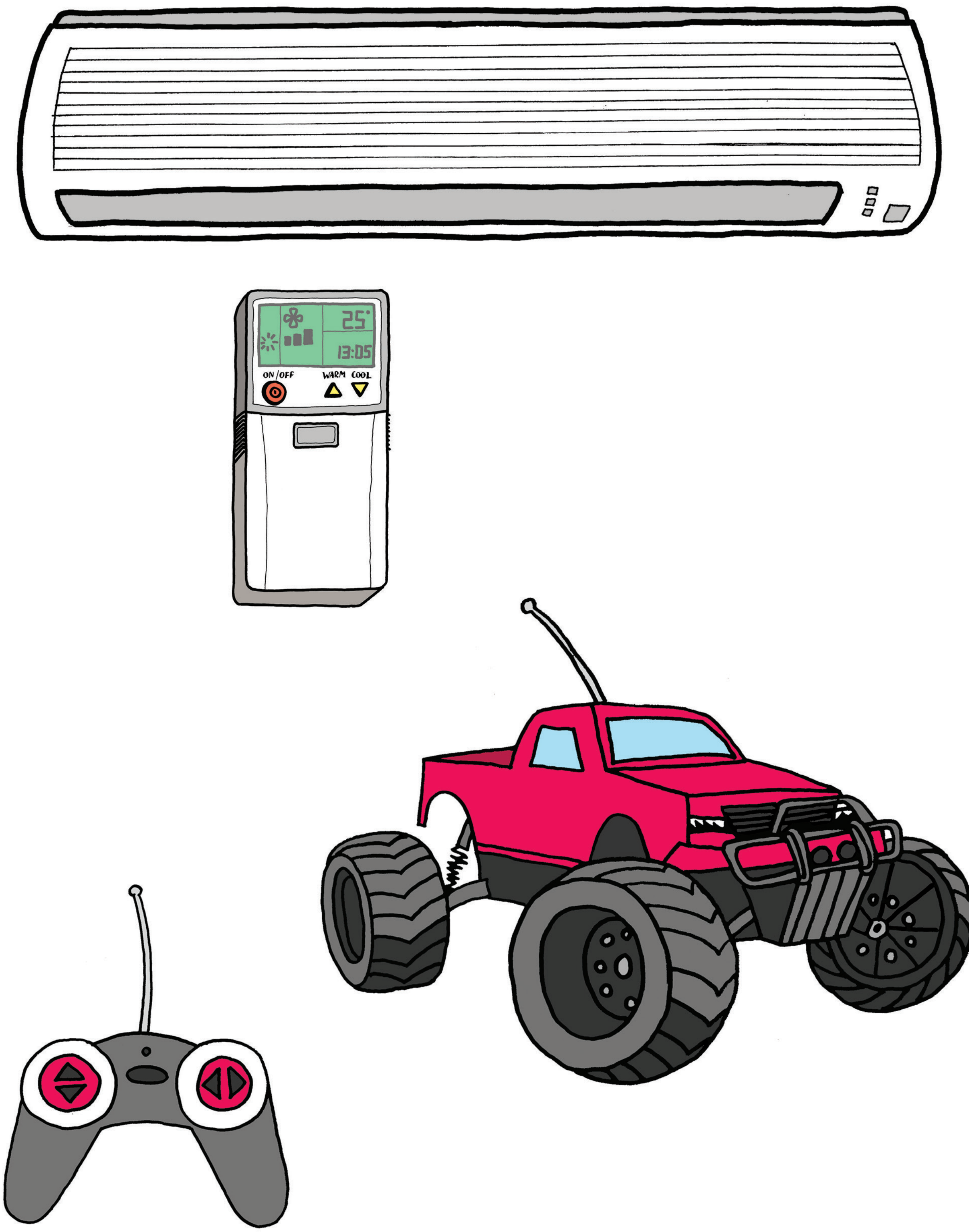
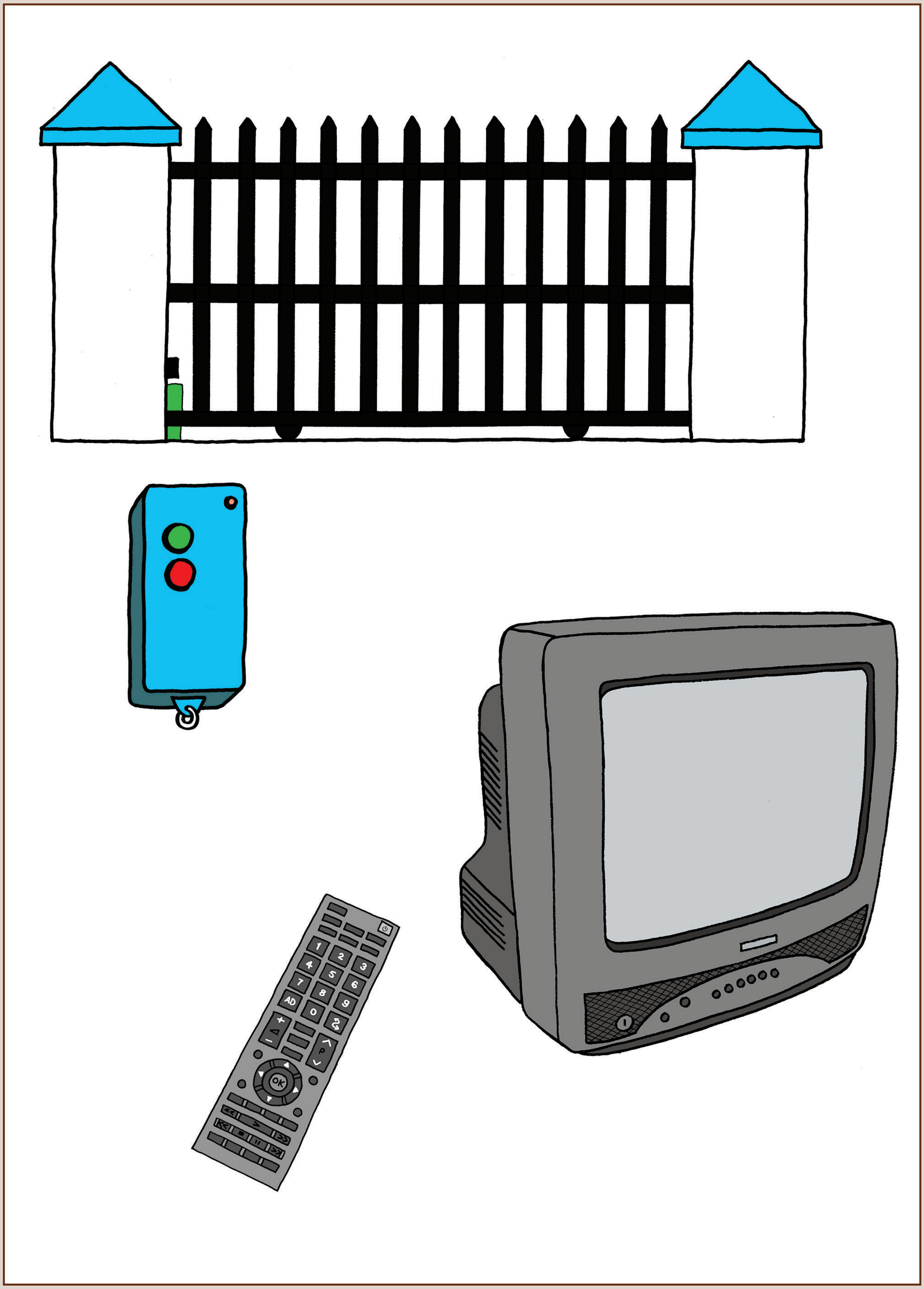



*Figure 1: Many everyday devices use electronic control circuits.*



# Week 1

## Situations where electronic circuits control electric circuits (30 minutes)

An **electronic circuit** is different from an electric circuit because it only uses a very small current, and because it uses electronic control devices such as thermistors, LDRs, diodes and transistors.

There are many household appliances that use **electronic circuits** to control electric circuits with bigger currents.

The following two devices are used inside the electric switchboard (or distribution board) of every building that is connected with electricity in a safe way.

- **Ordinary circuit breakers:**
  Shuts off a circuit (for example, the circuit supplying all the lights in a house) when the current becomes too big (if the current is too big for the thickness of wire used, the wire will overheat).

- **Residual-current circuit breakers:**
  Switches off the main power supply if it detects a leakage of power, such as when a person accidentally touches a "live" electrical wire or contact and the electricity is then conducted through his or her body. This device has to cut the current very quickly, otherwise the person can die due to electric shock. Therefore, it switches off the power even when it detects only a small amount of leakage of electrical current.


*Figure 2: An electrical distribution board with circuit breakers*

The following household appliances use electronic circuits to control them:

- **ovens:** to control the temperature,
- **radios and other music appliances:** to control the volume of the speakers,
- **some energy-saving lights:** to switch off automatically when there is enough natural light, and
- **kettles:** to switch off when the water boils.

1. Give two examples of situations or applications where electrical circuits are used. [1]
2. Give two examples of situations or applications where electronic circuits are used. [1]
3. Give three examples of situations or applications where electronic circuits and electric circuits are used together. [3]

[Total: 5]

## Investigate: A circuit with an input sensor, control knob, transistor and output device (15 minutes)

A *sensor* is a control device that can have a *variable effect*. A switch can only be open (infinitely large resistance) or closed (zero resistance), so a switch is not a sensor. Devices such as thermistors and LDRs can have different resistances, depending on the temperature or amount of light. They can therefore be used as sensors. A device that can generate a voltage, such as a photovoltaic cell, can also be used as a sensor. A sensor "senses" something such as temperature, or light, just as your body's senses do. A variable resistor is also a control device, but it is not a sensor, because it is a device for which the user can *set* the resistance.

The circuit for the fire alarm that you built in Chapter 15 can be used for different applications where a small input current from an input sensor has to switch on a circuit with a larger current for an output device. There is also a variable resistor so that the user can determine at what level of light or temperature (for example) the output device should be switched on or off.

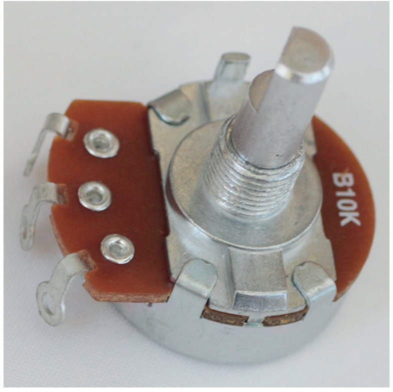



*Figure 3: The control knob of a stove plate is connected to a variable resistor. This controls the current through the heating element. The bigger the current, the hotter the plate will be.*

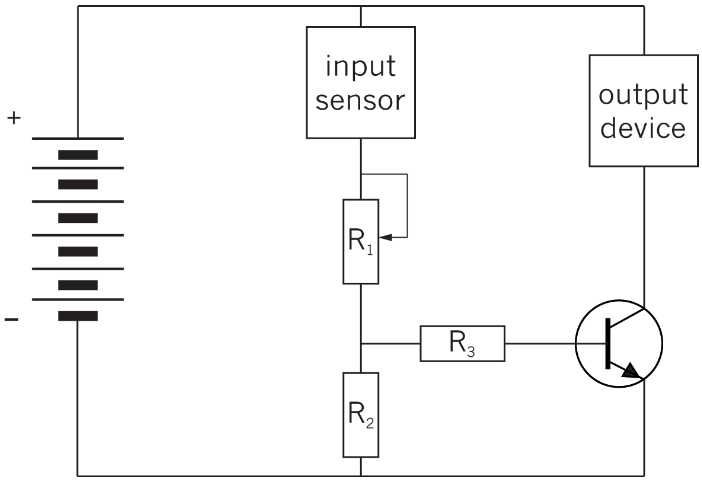



*Figure 4: The control circuit that you built in Chapter 15 for a fire alarm*

1. Name three input components that you know of.
2. Name three output devices that you know of.
3. Name a device that uses a control knob to set the level of something.

## Design brief and initial sketches (75 minutes)

### The scenario for the PAT

A kettle uses electricity at a rate 30 times higher than a normal light bulb. A lot of electricity can be saved if a kettle is used more effectively.

If a kettle keeps boiling without being switched off, it uses electricity unnecessarily. This leads to a waste of electricity.

If you drink your tea or coffee without cold milk, you do not want boiling hot water (100 °C), since it will burn you. So it is a waste of electricity and time to bring the water to boiling point (100 °C). Most of the time, a kettle only needs to heat water to a temperature of about 75 °C. If the kettle keeps heating the water to a temperature of 100 °C, it is a waste of electricity.

You will design and make an "energy-saving switch" for a kettle. The switch will be controlled by an electronic circuit so that the kettle will automatically switch off when the water reaches the required temperature. The electronic circuit will have a variable resistor so that the temperature at which the kettle will automatically switch off can be set by the user.

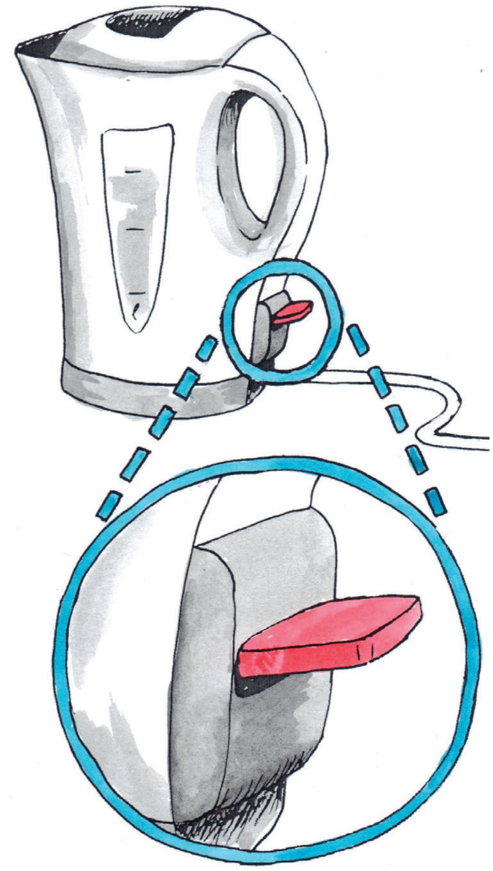


*Figure 5*

The drawings below show how an electric door lock works. This may give you useful ideas for your design of an energy-saving kettle switch.

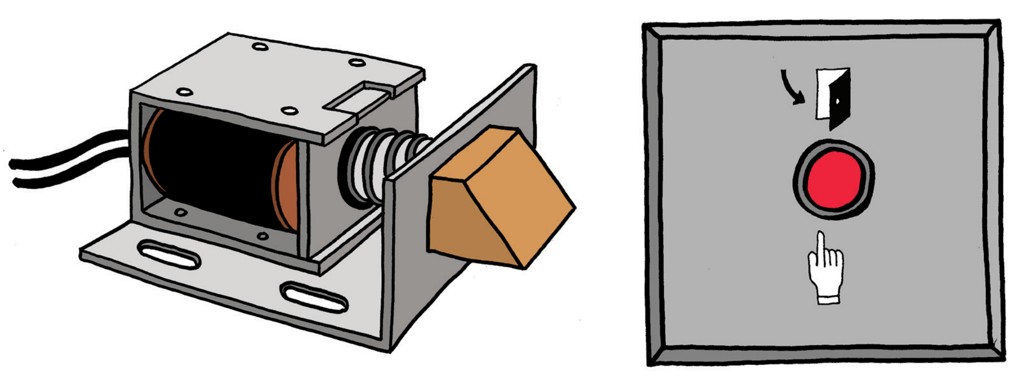


*Figure 6: An electric door lock*

Look at the brown part on the right-hand side of the lock mechanism above. This is the part that moves in or out to open or lock the door. This part is called a "latch".

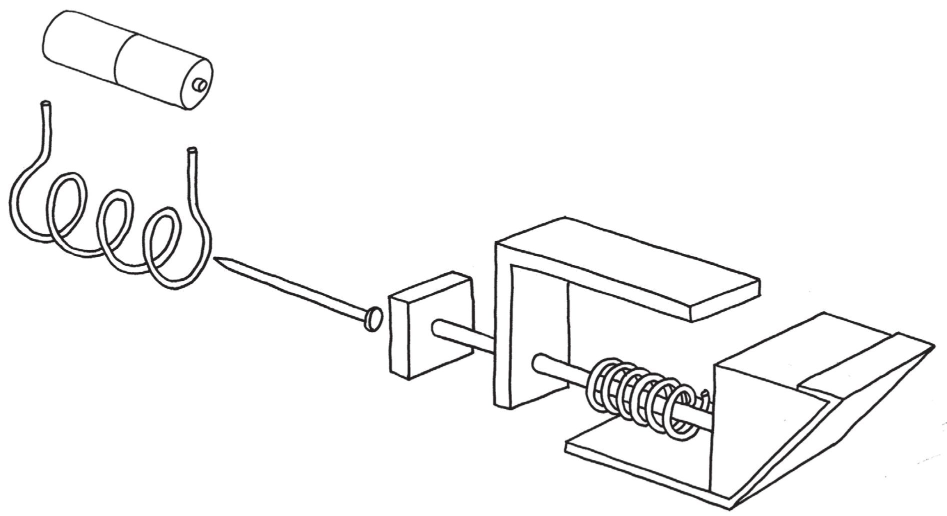


*Figure 7: A 3D assembly drawing of the parts inside an electric door lock*

**Design brief**

1. What is the purpose of the switch you will be designing?

   Hint: Think about how easy it is for people to do things, the impact on the environment, and costs involved. [2]

**Specifications**

2. What parts should the device have where the user must press or turn something by hand? [1]
3. Are there part(s) of the device that would sometimes be moved by the user, and other times be moved automatically? [1]
4. How should the moving parts of your switch work? For example, what should cause it to move one way, and what should cause it to move the other way? Use names for the different moving parts, as well as for the other parts that will make the moving parts move or stop them from moving. [2]
5. What type of electrical component can generate the automatic movement that your device has to perform? This component will be the output device in the control circuit on page 202.
6. Does your device need a container or supporting structure to keep all the parts together? What type of container or structure do you think will work well?

### Constraints

7. What property should the container of the device have, for safety reasons? Give the reason(s) as well. [1]
8. Make a time schedule showing how much time you have to design and make the product.

### Planning to make

9. Make a list of all the materials you will need.
10. Make a list of all the tools you will need.

### Design sketches

11. Make at least two rough sketches of your design. Use labels and notes to explain your design. If your second sketch is an improvement on your first sketch, keep the first sketch, but simply label the second sketch as "improved design". [5]

[Total: 12]

# Week 2

## Evaluate as a team: Learn from one another's designs to make a better design together (60 minutes)

1. Each team member should explain his or her design to the rest of the team, and the others should ask questions if they don't understand something.
2. After everyone has explained their designs, you should make a list of the advantages and disadvantages of all the designs.

**There is no such thing as a perfect design!** For example, you can make a complicated design that will work very well, but that will be expensive and difficult to build. Or you can make a simple and cheap design that works, but is not strong enough.

**Learn from the different designs that different people made**

Don't throw an idea away too quickly even if there is a problem with it. First sketch and explain it to the others. This idea can lead to another better idea. If everyone throws their ideas away too quickly, there will be no ideas on the table to work with. Design teams work well when they separate the work into two stages:

- First generate ideas, sketch and explain them, without anyone saying anything negative about the ideas.

If someone makes a negative remark at this stage, you should say *"Red flag! No negative remarks at this stage."*

- Once you have several ideas on the table, start thinking about how and whether the different ideas will work or not. Don't talk about "Mary's design" or "Sipho's design". Rather talk about "Design C" or "Design B". Once someone has put a design on the table, you talk about the design. You do not talk about the person. You evaluate the designs. You do not evaluate yourself or someone else.

Saying "Mary made a bad design" or "Sipho's is much better", for example, will hurt someone's feelings or make others feel proud or arrogant. If someone says "Mary's design ...", you should say *"Red flag! We call that Design C."*

3. Now combine different ideas from different designs into one better design. Your team will only succeed at this if you talk and sketch together "creatively". Being creative means "playing with ideas".

   To communicate well and to be creative, you have to make many rough sketches. Include labels and notes to help explain the sketches. [4]

4. Now each person should make their own sketches of the improved design that the team made together. Once again, show labels and notes to explain the sketches.

   Make at least two sketches, so that both the whole design and hidden detail can be seen. You might want to draw the design from different view points, or draw a few parts on their own. [4]

[Total: 8]

## Make individually: 2D working drawing and 3D drawings of your design (60 minutes)

Look back at Chapter 1 page 7 for an explanation of first-angle orthographic projections.

1. Make a 2D working drawing of your design in first-angle orthographic projection. It should be drawn to scale and show as much detail as possible. Show dimensions and the scale. Show all hidden details. [8]
2. Make an isometric drawing of your design to scale. Do not show the container or structural support for the inner parts of your design. Only show the inner parts. Do not show any hidden details, but choose your view point so that as much detail as possible is shown. Show the scale, but do not show the dimensions. [7]

[Total: 15]

## Homework: Planning to make and gathering materials

Make lists of the materials and tools you will need to build a model of your automatic kettle switch next week. You need to include the materials you will need to build the output device for the control circuit that you will later connect to your model of the switch. (Look back at your answer to question 5 on page 204.)

If there are any materials on your list that are not available at school, gather waste materials that you can use instead and bring it to school next week. *If you do not do this, you won't be able to build a model of your design.*

# Week 3

## Make and test your prototype of the switch

**(120 minutes)**

You should work only individually in this section, with your own model. There is one exception, namely question 4.(a). For that question, you should work as a team to test the control circuit before you connect it to your model. After that you should return to working individually.

Designers and engineers usually make many **prototypes** before the design is good enough to start manufacturing and selling it. Each prototype is an attempt to improve on the previous one.

1. Work alone to build a model of your design for the switch. A model of a new design is called a **prototype**.
2. Work alone to build the output device for the control circuit that you will later connect to your switch.
3. Test your model with a simple circuit consisting of a battery and the electric output device that you made.
4. Test your model by connecting it to the control circuit that you made in Chapter 15, but this time connect a 9 V battery to the circuit, instead of the 6 V battery you used in Chapter 15. The output device needs a bigger potential difference than the buzzer in Chapter 15 did.
   (a) Before you connect the control circuit to any model, your team should test the control circuit as you did before (see page 192), because some of the connections may have come loose.
   (b) To test your automatic kettle switch, you can use a thumb tack pressed into an eraser that you heat by rubbing it on a piece of wood or plastic for a minute.
   (c) If you were not able to build a control circuit successfully in Chapter 15, you can use the simple circuit discussed in question 3 above to test your model of the switch.
5. You will probably find that your model does not work the first time you test it. This is normal! Most new things that people design don't work the first time they test it. Try to find out what's wrong, and then go back and fix it before you test it again.

Your teacher will give you marks for the following:

- You brought all the materials needed to make a model of your design. [2]
- You accurately made the model according to your design drawings. [8]
- You successfully built the electric output device. [2]
- You connected your model to the simple circuit with the output device, and used a good method to test it. [1]
- After you tested your model for the first time, you made a list of all the possible reasons that your model is not working or why it is not working well. [2]
- You used the list to fix or improve your model. [2]
- You tested your model again, writing down the problems, and going back and fixing or improving your model until it worked, at least one more time. [4]
- Your model worked, or you wrote a good explanation and made sketches of what you still need to change on your model to make it work. [4]

[Total: 25]

You need to keep a record of all your testing and improvements on your model, otherwise you will not get marks for that work.

# Week 4

## Present your design process and final prototypes

Your team will give a presentation of your project later this week. The presentation should be between three and five minutes long. Each member of your team should do a part of the presentation. The other learners in the class may ask you questions after your presentation.

Your presentation should be mostly about the design process that you followed to design, make and improve your prototypes.

### Team meeting: Prepare your presentations (30 minutes)

1. Decide which part of the presentation each of you will do. Write it down. [1]
2. Decide in what order you will give the different parts of the presentation. Who will talk first, and who will talk next?

   Write the parts of the presentation in the order that you will do them, and write who will do which part. [1]
3. For homework, you should practise your part of the presentation.

[Total: 2]

### Giving the presentations (90 minutes)

Your teacher will look at the following to give you marks for your part of your team's presentation:

- You were well prepared for your presentation. [2]
- You explained how you made progress during the design process. [2]
- You looked at your audience and spoke clearly. [1]

[Total: 5]

## An alternative to the kettle switch project: Designing and building a circuit continuity tester

Your teacher may decide to let you do the following project instead of designing and building an automatic kettle switch.

Often when people have to connect wires in electric circuits, there are so many wires that it is difficult to know which two wire ends are of the same wire.

It would help to have a device that shows whether two wire ends are connected or not. This is what a “circuit continuity tester” does.

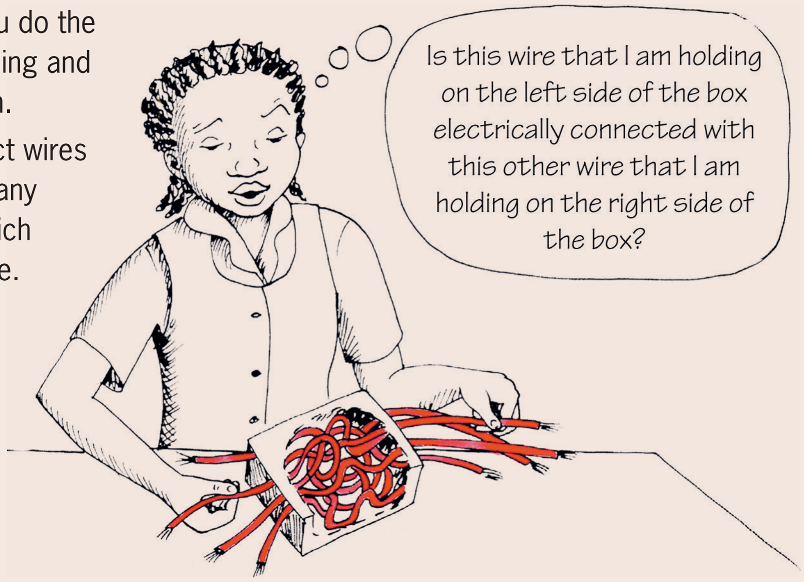



*Figure 8*

A circuit continuity tester is actually an open circuit. The circuit can only be closed by the two wire ends that you are testing. Use the two test leads of the circuit continuity tester to touch the two wire ends that you want to test. If there is a path for current to be conducted between the two wire ends, this will complete the circuit and a light or a buzzer on the circuit continuity tester will be activated.

**Safety warning:**
First switch off the power supply before you do a test such as this one.

Note that a circuit continuity tester cannot tell you whether the two wire ends are of the same wire. It can only tell you whether there is a path for current to be conducted between the two wire ends, in other words whether the two wire ends are electrically connected. But if you know that there are no splitting or joining of wires in between the two wire ends, then the wire ends can only be electrically connected if they are of the same wire.

If you design and build a circuit continuity tester as your project, think about the following:

- It should be easy to let the test leads of the circuit continuity tester make proper electrical contact with the wire ends.
- The tester should be small.
- The tester should be protected from shocks, for example if it gets dropped.
- The tester should be protected from water, since water can cause a short circuit.

A few ideas for building a circuit continuity tester are shown in the photos below.

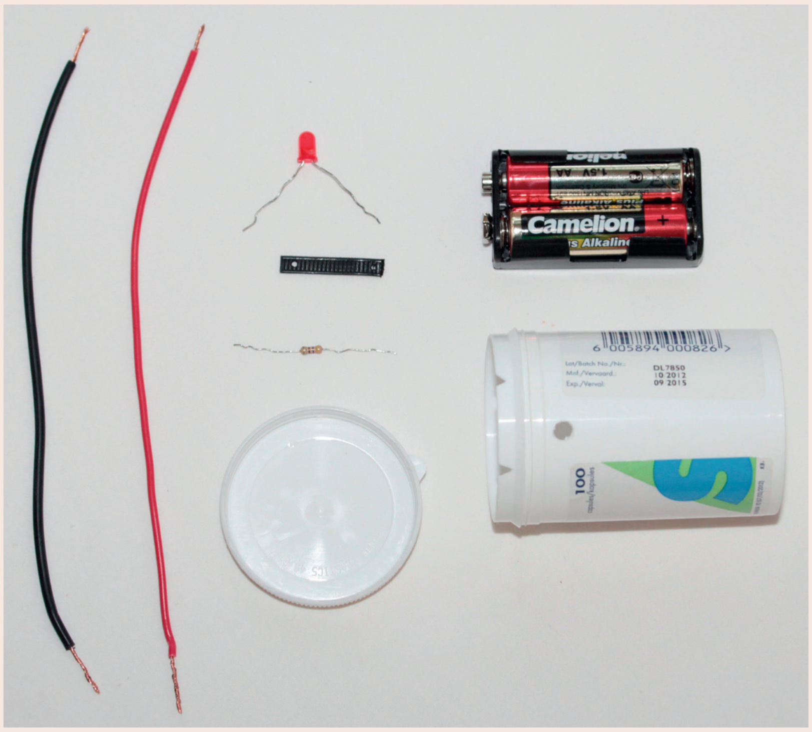



*Figure 9*

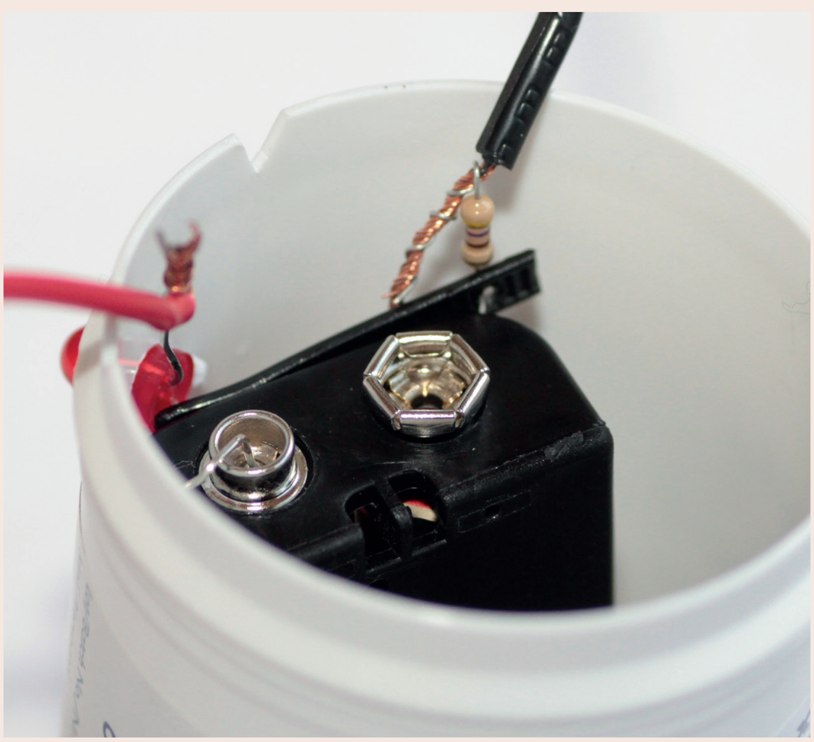


*Figure 10*